Engineering Mechanics 1 · Unit_04 · UNIT 4

DRAFT — UE26ME142A · Unit_04 — UNIT 4 · Complete Unit Master

Notes-first study master · source-cited lecture explanations and compact page takeaways

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the edited lesson notes by topic. Evidence chips point to transcript intervals and slide pages. The complete source transcripts, page text, formula candidates, and image locators are preserved in sibling JSON datasets. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

Contents

1. 1 · UE26ME142A · Unit_04: UNIT 4
2. 2 · CLASS 46 Introduction to Moment of Inertia
3. 3 · CLASS 38 Moment of Inertia_ Numerical
4. 4 · CLASS 39 Moment of Inertia_ Numerical
5. 5 · CLASS 40 Moment of Inertia_ Numerical
6. 6 · CLASS 41 Moment of Inertia_ Numerical
7. 7 · CLASS 51.1 Moment of Inertia_ Numerical
8. 8 · CLASS 51.2 Moment of Inertia_ Numerical
9. 9 · CLASS 51.2 Moment of Inertia_ Numerical
10. 10 · CLASS 47 Friction_ Numerical
11. 11 · CLASS 48 Friction_ Numerical
12. 12 · CLASS 56.1 Friction_ Numerical
13. 13 · CLASS 56.2 Friction_ Numerical
14. 14 · Source slide deck: Moment of Inertia
15. 15 · Source slide deck: UE20CV101 UNIT 4 SESSION 6
16. 16 · Source slide deck: UE20CV101 UNIT 4 SESSION 11
17. 17 · Source slide deck: UE20CV101 UNIT 4 SESSION 12
18. 18 · Source slide deck: UE26ME142A Unit 4 class 53 Lecture PPT
19. 19 · Source slide deck: UE20CV101 UNIT 5 SESSION 2
20. 20 · Source slide deck: UE20CV101 UNIT 5 SESSION 3
21. 21 · Source slide deck: UE20CV101 UNIT 5 SESSION 6
22. Appendix A · Symbol glossary
23. Appendix B · Sources, coverage and gaps
24. Appendix C · Revision and retrieval practice

M1

Unit_04 · UNIT 4

1 UE26ME142A · Unit_04: UNIT 4

This unit master organizes source material for Engineering Mechanics 1, Unit_04, officially titled UNIT 4. It includes 12 video transcripts and 15 source slide decks, with page-level text and rendered images where available. This draft preserves source wording and page order; it does not claim a complete human subject review or full audiovisual review. EV0000001

SHORT-NOTE RECAP

Unit map · UE26ME142A · Unit_04: UNIT 4

Official catalogue entry: UE26ME142A, Unit_04 (UNIT 4). Bucket folder present: yes. Videos: 12; slide decks: 15.

EV0000001

2 CLASS 46 Introduction to Moment of Inertia

Study focus: CLASS 46 Introduction to Moment of Inertia. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000002 · EV0000003

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

Moment of inertia is an object's resistance to bending at the rotation, and it is also called as second moment of an area.

Key details

- The concept of first moment of an area we discussed in locating the centroid of a lamina.
- It's nothing but second moment of an area.
- That's why moment of inertia I will be equal to distance square d square into area A.
- And it is also.

- A purely a mathematical property of an area and itself doesn't have any physical significance.
- Rectangular moment of inertia is nothing but moment of inertia phenamina about any two mutually perpendicular axes.
- Because X and Y we know they'll be always mutually perpendicular to each other.
- Then moment of inertia about an axis which is perpendicular to the plane of an area or a lamina that we call it as a.
- Polar moment of inertia.
- As we know, z axis is always perpendicular to xy plane or a plane of a lamina.
- You can see here.
- It is in the XY plane.
- about the x axis then about the y axis.
- Let the area of this element let it be d.
- I want to calculate.
- Since area is very small, the centroid of this area will be the coordinates of the centroid are going to be x, y.
- Distance is y, so that we know.
- This is moment of inusia.
- Since we are talking about the smaller element, it.
- We go, we are going to take it as DIX.
- This is this stands for moment of inertia of the elemental area about x-axis.
- Again it is distance square into area.
- Distance here is x.
- Once we know moment of inertia of the element about x and about y axis.
- That is ix is equals to integration.
- integral of y square into d a and iy is integral of x square into d a. i x stands for inertia of the lamina about x axis iy stands for inertia of the lamina about y axis since x and y are perpendicular to each other i x and i y.
- Why we call them as rectangular moment of inertia.
- As I said before, moment of inertia about z axis we call them as polar moment of inertia.
- Since there is perpendicular to the plane of the lamina.
- Nothing but perpendicular to XY plane, we can't see that.
- Just remember that is passing through point O.
- Then how to find inertia about the Z axis passing through O?
- We know moment of inertia is distance square into area.
- This is the moment of the smaller element about the z axis.
- Next, I want to find the moment of inertia of this entire lamina about.
- Z-axis, so just differentiate integrate that.
- This is called as polar moment of inertia.

Example or application from the lecture

- Example like X and Y axis.
- We consider moment of inertia about z axis as the polar moment of inertia.
- To explain this, I am going to consider this figure.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000002 · EV0000003 · EV0000004 · +1 source units in pack.json

Core explanation · 05:00–09:59

There is a relationship between polar moment of inertia and rectangular moment of inertia according to perpendicular axis theorem.

Key details

- According to perpendicular axis theorem, moment of inertia about a z-axis, which is perpendicular to the plane of the lamina, will be always equals to.
- Sum of the moment of inertia about the two mutually perpendicular axes passing through the same point O and lying in the same plane of the lamina.
- Nothing but IZ is equals to Ix plus I.
- The condition is.
- X, Y, Z should be passing to the same point.
- If you look at this, there is a right angle triangle.
- Once you substitute here, i z will be equals to integral of x square into d a plus integral of y square into d a.
- From here we know integral.
- integral of x squaring to d is i y and integral of y squaring to d a is i x so iz will be equals to i x plus i y.
- Once we know the rectangular moment of inertia about any point, like O R it can be anything, then we can find polar moment of inertia about the same point just by adding Ix and Iy.
- We are going to get this IZ.
- An element whose coordinate is negative contributes as much as the moment of inertia as does an equal element with a positive coordinate of the same magnitude.
- Since these two are identical, assume.
- The centroid will be some area.
- Since this point centroid is on the right side of the axis of symmetry, it will be positive.
- But since we are talking about the distance square, even though it is on the left side of the axis, even though it is minus x, distance square of the minus 6 will be positive.
- It's a continuation of the same thing.
- Whereas the first moment of area can be positive, negative, or it can be zero, whereas the second moment of an area will be always positive.
- As I said before.
- First moment of an area, we use this concept in locating the centroid.
- Because first moment of an area is area into distance, if the x coordinate is negative, then.
- The first moment of area can be negative.
- Few times it will be zero.
- When the coordinate is positive, first moment of area can be positive.
- Since here it is distance square, so it doesn't matter whether that is positive or negative.
- Next is radius of guidation.
- The total area of the lamina let it be A.

- Then Ix stands for moment of inertia of the lamina about x-axis.
- Iy stands for moment of inertia of a lamina above y-axis.
- X bar and y bar are the coordinates of the centroid.
- This is what we know in this particular case.
- We know area, we know Ix, Iy, X bar, and Y bar.
- We are going to visualize this area as a narrow rectangular strip.
- Just area remains the same area as.
- The same area here, and the moment of inertia of this narrow rectangular strip, which is parallel to x-axis, that is the same as that of the case one.
- The inertia in this case.
- Is also Ix.
- Then how to find kx?

Example or application from the lecture

- To prove that, I'm going to consider here.
- Next, say there is a lamina like this, and this is the y axis given in the cushion.
- To explain this, I am going to consider this rectangle.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000006 · EV0000007 · EV0000008 · +1 source units in pack.json

Core explanation · 10:00–14:59

We know from the definition of moment of inertia, moment of inertia is distance square into area.

Key details

- Since we know Ix from here and a from this case, now kx can be written as under root of ix by area.
- Area as a narrow rectangular strip which is parallel to y axis.
- Then inertia of this narrow strip about y-axis is also the same as that of.
- This case i y.
- Ky will be equal to under root of iy by total area A.
- And in this case also, momentum inertia of this area about.
- The point about the z-axis passing through O will be the same as that of the moment of finesse about the z axis passing through O in this case.
- Ky is under root of IY by A, Kz is equals to under root of IZ by area.
- When we know the value of moment of inertia about x-axis, then and we know total area, then we can find radius of variation about x-axis.
- And it is a measure of distribution of the area from the axis in the cushion.
- And as we know here, I z is equals to Ix plus I.
- I z we know kz square into area, that will be equals to kx square into area plus ky square into area.
- Since area is.
- Thus, the square of the radius of guaration about the polar axis equals to the sum of the squares of the radii of variation about the two corresponding regular rectangular axis.

- It's also called as transfer of axis.
- According to parallel axis theorem, moment of inertia about any axis will be always equal to sum of the moment of inertia about parallel centroidal axis plus product of the area and square of the distance between the parallel.
- This is the statement of the.
- Parallel axis theorem mark you can write it as dh.
- That is the distance between the two x axis.
- Like the area of this lamina let it be capital A.
- A is the area of the lamina.
- This is x-axis and this is.
- Let C be the centroid of this lamina.
- Then let it be at a distance of dx with respect to.
- We know how to find DX.

Example or application from the lecture

- If you consider moment of inertia about any horizontal axis, so I'm going to write it as ix will be always equals to moment of inertia about centroidal x axis.
- To prove this, I am going to consider there is a lamina like this.
- Next, consider this is in the xy plane like this.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000010 · EV0000011 · EV0000012 · +1 source units in pack.json

Core explanation · 15:00–19:59

Next, I am going to consider one smaller elemental area within this lamina.

Key details

- The shaded region, let it be the smaller elemental area, let d be the area of this element.
- Let it be at a distance of 10 with respect to.
- The reference line here, the reference line is centroidal x-axis, so let it be y naught.
- Smaller element above the x axis.
- What is moment of inertia?
- Moment of inertia is second moment of an area, that is distance square into area.
- Distance from ear to ear is y naught plus dx.
- Whole square into area is dA.
- This is the moment of inertia of the smaller element about the x-axis.
- It's going to be a square.
- Y not square into DA.
- What is this?
- It is dis it is in the form of distance square into area.
- What is distance square into area?
- Distance square into area is second moment of an area.
- Nothing but moment of inertia.

- Since the distances with respect to centroidal x axis and area is here, it is moment of inertia bar centroidal x axis.
- Second term is integral of dx square into dm.
- This is constant, dx is constant.
- What is d?
- Integral of d a integral of d a is total area.
- The last term is two into integral of y naug into dx into d.
- This will be equal to here.
- Dx is constant.
- Here I am going to divide it by total area and multiply with total area.
- It doesn't make any difference.
- Since the central is on the reference line, this will be equal to 0.
- This entire term will be 0.
- So we are going to get IX as Ix is equal to first term integral of y naught square into d is i bar x.
- I bar X plus this is dx square into A and this is 0.
- Nothing but moment of inertia about any axis will be always equal to.
- Sum of the moment of inertia about parallel centroidal axis plus product of the area and square of the distance between the two parallel axes.
- Here you have to remember two points to apply the parallel axis theorem.
- The first condition is the axis.
- And the centroidal axis both should be parallel to each other.
- And to use this formula, one of the axes should be the centroidal axis.
- Then only we can apply parallel axis theorem to find the moment of inertia above any axis.
- that's it for today's class.

Example or application from the lecture

- Next, consider each term.
- First, I am going to consider the term integral of.
- Next consider the second term.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000014 · EV0000015 · EV0000016 · +1 source units in pack.json

3 CLASS 38 Moment of Inertia_ Numerical

Study focus: CLASS 38 Moment of Inertia_ Numerical. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000018 · EV0000019

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

Question A slash one.

Key details

- Determine the moment of inertia of a rectangular area about the X and Y axis, and also find the polar moment of inertia about the point O.
- In this question, they have asked us to find inertia of this.
- Inertia of this rectangular lamina about this x-axis, then about the y-axis.
- After that we have to find polar moment of inertia about the point O.
- Polar moment of inertia about any point will be equal to some of the moment of inertia about the x and y axis passing through the same point, that is according to perpendicular axis.
- Then how to find inertia of this rectangular area about this axis.
- When you think of rectangle, you may remember the formula about the bottommost member as BH cube by 12 and about the centroidal x axis as bh cube by 3.
- If you look at this x-axis, it is not coinciding with the bottommost member.
- But what about is it a centroidal x-axis?
- Where will be the centroidal x-axis?
- in a rectangle.
- We know the y bar formula of a rectangle that is h by 2.
- Here what is h? h is 2 h by 3 plus h by 3, it is h itself.
- If we draw a horizontal line, this will be the centroidal x axis.
- That's why it is called as x naught.
- We don't have a standard formula.
- Then how to find inertia about this axis, x axis.
- Whenever we don't have a standard formula, we have to apply the parallel axis theorem.
- Inertia about any axis is equal to inertia about.
- Parallel centroidal axis plus area into distance square.
- This is the statement.
- Now I am considered I bar x because.
- This is horizontal, so centroidal axis also should be horizontal.
- Hence it will become I bar X.
- Next, how to find the value of I bar X then area and D.
- This is from the formulas.
- A is nothing but area of a rectangle.
- Here what is area of a rectangle?
- Base into height.
- What is this width or a base?
- It's it is B and the height is H.
- Then what is D?
- D stands for the distance between these two parallel lines.
- Nothing but X0 and X axis.
- And what is this distance?
- From here it is h by 2 minus h by 3.
- H by 2 minus h by 3 if you simplify and get it as h by 6.

- After substituting in this, simplify this, we will get the value of ixs bh cube by 9.
- This is the moment of inertia of a rectangle about the x axis.
- Next is we need to find moment of inertia of a rectangular area about the y axis.
- If you remember the formulas again, we know inertia about the leftmost member is Hbq by 1.
- Inertia about the centroidal y-axis is HBQ by 3.
- At the same time, is it a centroidal y-axis?
- This width of the rectangle is B.
- Where will be the centroidal y-axis in a rectangle?

Example or application from the lecture

- Let's start with finding Ix, that is moment of inertia of a rectangular area about the x axis.
- Let's check that.
- If you remember the basic formulas, when we consider a rectangle, inertia of this rectangle about its centroidal x axis is BH cube by 2 y.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000018 · EV0000019 · EV0000020 · +1 source units in pack.json

Core explanation · 05:00–09:59

At a distance of B by 2, if you draw a vertical line, this will be the central y axis.

Key details

- Since we don't have any formula about this y-axis, we have to apply parallel axis theorem to find inertia of this rectangular.
- lamina about the y axis.
- Inertia about any line that is i y is equal to inertia about parallel centroidal axis that is i bar y plus area into distance square.
- Always remember the statement of the parallel axis theorem.
- Next, what is this I bar Y inertia about centroidal y-axis?
- Remember the formulas.
- From the formulas we know inertia about centroidal y axis, that is this value, is HBQ by 12.
- D stands for the distance between the two parallel axes, that is y and y naught axis.
- What is the distance?
- That is b by 2 minus b by 4.
- You will get it as b by 4 itself.
- Then substitute the value of i bar y, area and distance in the equation 2 and simplify we will get the value of i y.
- The last part of this question is polar moment of inertia about the point O.
- As I said before, IZ about O is equal to Ix plus Iy.
- From the previous step Ix is BH Q by 9, substitute that here and Iy is 7 HBQ by 48.
- Once we simplify this, we will get the value of IZ as BH in.
- Ah into H square by 9 plus 7 B square by 48.
- This is the polar moment of inertia for a rectangular lamina about the point O.

- A slash 3.
- There is a narrow rectangular strip as an area of 300 mm square and its moment of inertia about the y axis is 35 to 10 power 3 mm to the power of 4.
- Open a close approximation to the polar radius of guardation about the point O.
- How to find polar radius of guardation?
- Polar radius of guaration is nothing but Kz.
- For Kz we have two formulas.
- Kz is equals to under root of IZ by A.
- That we can use when we know the polar moment of inertia about the.
- When we know the polar moment of inertia.
- This one more formula that is.
- Kz is equals to under root of kx square plus ky square.
- This formula we can use when we know the value of radius of guaration about x axis and radius of guaration about the y axis.
- Next, look at the figure.
- That is nothing but radius of guarantee the x axis.
- Since we know iy and area, we can find the value of radius of guaration about the y axis, that is ky.
- Ky is equals to under root of iy by total area A.
- Once we simplify this, we will get the value of Ky radius of variation about y-axis.
- We know radius of variation about x axis and radius of variation about y ax.

Example or application from the lecture

- Let's get into the next question.
- Before that, let's tabulate what is given in the cursing.
- The area is given as 300 mm square, consider that.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000022 · EV0000023 · EV0000024 · +1 source units in pack.json

Core explanation · 10:00–14:59

Simplify this, we will get the value of radius of garation about the point O as 18.48 mm.

Key details

- This is the final answer.
- The question is determine the ratio B by H such that Ix is equal to Iy for the area of the isoscerus triangles.
- Yet B is.
- This base of this triangle and edge is the height.
- What is Ix?
- Ix is moment of inertia about the x-axis, and what is Iy?
- I is moment of inertia of this triangle about the y axis.
- When we have a triangle, assume a right angle triangle or any other triangle.

- Moment of inertia about its base and centroidal x-axis and about the axis passing through its apex remains the same, irrespective of whether that is a right angle triangle or any other triangle like this.
- About this line is always about its base will be always BHQ by 12 about the line horizontal line passing through the apex is BHQ by 4.
- Irrespective of the triangle these two will remain the same.
- Moment of inertia this triangle about its ax, that is i will be equal to bh q by 4.
- I am going to use this formula.
- Then in this what is B?
- B is nothing but B itself, that is B by 2 plus B by 2, nothing but B, and H, what is H?
- Height of the triangle here also it is H.
- Next, I need to find IY.
- When we think of initially about y-axis, we should have only the right angle triangles.
- Here, this is one right angle triangle and this is also one more right angle triangle.
- Since these two are similar, iy will be equal to the formula is hbqed by 12.
- Next, since these two are similar, I can take it as 2 times that of 1.
- Because moment of inertia of this triangle about this line will be equal to moment of inertia of this triangle about the same line or same axis y axis.
- Next, if we have to get into the substitution part.
- Here what is h?
- H is this.
- That is nothing but H itself.
- Next, what is B?
- B stands for the base of the triangle.
- Since we have considered two triangles, the base should be taken as B by 2.
- Next, simplify this.
- Once you simplify you will get it as h bqed by forty-eight.
- Since Ix is equal to Iy, H B H cube by 4 will be equal to H B cube by 48.
- Solve it then.
- We need to find the ratio of B by H.
- The ratio will be equal to under root of 12.
- If you come across any doubts, you can post your queries in the PESU forum.

Example or application from the lecture

- Let's start with the next question A slash 4.
- And what is given here is Ix is equal to Iy.
- Try to solve the problems given in the assignment as well as in the question bank.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000026 · EV0000027 · EV0000028 · +1 source units in pack.json

4 CLASS 39 Moment of Inertia_ Numerical

Study focus: CLASS 39 Moment of Inertia_ Numerical. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000030 · EV0000031

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

The first question is A slash six Determine polar moments of inertia of a semicircular area about points A and B.

Key details

- What is polar moment of inertia?
- It's a moment of inertia about Z axis passing through that particular point.
- Since we are talking about polar moment of inertia about A, it is the moment of inertia about Z-axis passing through point A.
- And remember that z axis will be always normal to the plane of the paper.
- Ah the point A itself, that's the point to which the Z axis is passing, and that is normal to this plane of this semi-circular area.
- When we don't have a standard formula, when we don't have a standard formula, we have to apply the parallel axis theorem.
- Inertia about any axis is equal to inertia about parallels intoidal axis plus C area into distance square.
- This is the statement which we know.
- Since we are talking about the Z axis, here the centroidal axis also should be the centroidal Z axis.
- Next, where will be the centroidal?
- Centroidal Z axis will be passing through C and that will be normal to the plane of the paper.
- Remember the basic formulas whenever we have a semicircle.
- First locate the center.
- This is the center of a semicircle.
- Then draw the axis of symmetry.
- Then the distance.
- From the centroid to the center on the axis of symmetry is 4 r by 3 pi.
- This is 4 r by 3 pi, we will get this point C.
- That is nothing but z-axis passing through C.
- For that, remember I z will be equals to Ix plus Iy, since here it is the Z axis passing to the centroid, it will be a centroidal z axis.
- That will be equal to I bar X plus I bar.
- Y where I bar x is moment of inertia about centroidal x axis and i bar y is moment of inertia about centroidal y axis.
- Moment of inertia about a centroidal z axis.

- Then what is i bar x?
- I bar x we know from the formulas it is 0.11 into r to the power of 4.
- Then what is i bar y?
- That is pi r to the power of 4 by 8.
- And these two.
- We are going to get high bar Z.
- Remember, this is the moment of inertia about the centroidal Z axis.
- Once we know moment of inertia about centroidal z axis, we can find moment of inertia about z axis about any other point.
- Z axis is passing through point A, we have to find inertia about that Z axis.
- How to find that?
- That will be equal to inertia about centroidal Z axis, that is I bar Z plus.
- Area into distance square.
- This is the statement.
- Then what is I bar Z?
- We know moment of inertia about centroidal z axis is 0.503 R to the power of 4.
- Plus, area is area of a semicircle, that is pi r square by 2.
- Then what is D?
- D is the distance between the two parallel axes.
- Here, one axis is passing through A and one more axis is passing through C.
- Then how to find this distance AC?
- To find this distance A C, look at this.
- This is C which is at a distance of four r by three pi.
- Then we need to find this distance A C.
- This A C will be equal to under root of AO square plus C O square because C O A is a right angle triangle.
- Then what is A C?
- A C is equal to under root of R square.
- That is AO plus 4 R by 3 pi whole square.
- We will get the value of distance between the centroid and Z axis and the Z axis passing through A.
- And Area we know.
- Substitute the value of area, then the distance and I bar z in the.
- Equation 1 and simplify we are going to get the polar moment of inertia about the point A.
- That is 2.35 into R to the power of 4.
- This is nothing but moment of inertia about z axis passing through A.

Example or application from the lecture

- Now let's start with the first point, polar moment of inertia of this semicircular area, about a point A.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000030 · EV0000031 · EV0000032 · +1 source units in pack.json

Core explanation · 05:00–09:59

Next, we need to find moment of inertia, polar moment of inertia about point B.

Key details

- Inertia about polar moment of inertia about point B will be equals to moment of inertia about centroidal z axis, that is I bar Z plus area into distance square.
- I bar z we know from the previous case.
- This is 0.503 R to the power of 4, substitute that.
- Then area is area of the semicircular lamina that is pi r square by 2.
- D is the distance between the centroidal Z axis and the Z axis passing through B.
- Nothing but this C B distance.
- Then how to find this C B?
- C B is equals to year to year OB is R, R minus O C is 4 R by 3 pi.
- Once we know the distance, area, then I bar said, substitute these three in the equation 2.
- And we have to simplify this.
- Then we will get the polar moment of inertia about the point B.
- This nothing but moment of inertia of this semicircular area about the z-axis passing through B.
- That is 1.023 into r to the power of 4.
- This is the answer.
- Next question A slash 7 determine the moment of inertia of a quarter circular area about the y axis.
- Then we have to apply parallaxis theorem.
- Inertia about y axis is equals to initiate about centroidal y axis plus area into distance square.
- This is the statement.
- Then what is I bar y?
- For a quarter circle, we know from the formulas I bar y is inertia about centroidal y axis that is 0.055 into r to the power of 4.
- Remember the formula?
- Substitute that i bar y value.
- Since e r is a, it is going to be 0.055 a to the power of 4.
- Okay, next we have A.
- A is area of this quarter circle that is pi r square by 4.
- Since r is a, it is it will be pi a square by 4.
- D is the.
- Distance between the y axis and the centroidal y axis.
- For that, where will be the centroid with respect to the radial line?
- It will be at a distance of 4 r by 3 pi.
- But what is the distance here?
- Distance is this distance we want.
- That is the distance between y ax and y naught axis.
- Then how to find this distance?
- This distance will be equal to a by 2 plus a minus of 4r by 3 pi.

- Here R is A, so substitute R S A in this we will get the value of D.
- It is 1.07 double phi into a.
- Substitute the value of I bar y area then D in this equation one, then simplify this, we will get the value of I as 0.96368 to the power of 4.
- This is the moment of inertia of this quarter circle about the about y axis.
- This is the next question A slash 19.
- Determine moment of inertia of the shaded area about the X axis.
- Look at this shaded area.
- Shaded area is only this.
- We have to find inertia of this shaded area above x axis.
- First thing, do we have any standard formula for the shaded area?
- We don't have.
- Then how to get this shaded area?
- If you observe this figure carefully.
- This is a quarter circle of radius A in that this triangle has been cut or removed of base A by 2 and the height A.

Example or application from the lecture

- Let's consider that again we are going to apply the parallelaxis theorem.
- Y axis is given here, which is at a distance of A plus A by 2 with respect to this origin of this.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000034 · EV0000035 · EV0000036 · +1 source units in pack.json

Core explanation · 10:00–14:59

Next, how to find inertia of the shaded area?

Key details

- Moment of inertia of the shaded area about x axis will be equal to.
- Moment of inertia of this entire quarter circle about x axis minus in that what do we need to detect?
- Inertia of a triangle about the same x axis.
- I x is equal to moment of inertia of a quarter circular lamina about x axis minus moment of inertia of a triangle about the same x axis.
- Next, what is moment of inertia of this quarter circular area about x axis?
- This x-axis is coinciding with the radial line.
- Here r is a, so it will be pi a to the power of 4 by 16.
- We are done with the quarter circle.
- Next come to the triangle.
- What is the moment of inertia of a triangle about its base?
- The standard formula is BHQ by 12.
- Here what is base A by 2, and what is H A.
- Substitute in this formula, we are going to get the value of Ix.

- Radius is a and base of the triangle is A by 2, and the height of the triangle is A.
- Substituting this the above equation, it's going to be pi A to the power of 4 by 16 minus basis.
- A by 2, take that as it is.
- H is J 8 cube by 12.
- Simplify this, we will get the value of Ix is 0.154 a to the power of 4.
- Remember, this is the moment of inertia of a shaded area about the x axis.
- That's it for today's class.
- If you come acros any doubts, you can post your queries in the PESU forum.

Example or application from the lecture

- That's what is given here.
- That's what is given here.
- Try to solve the problems given in the assignment as well as in the question bank.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000038 · EV0000039 · EV0000040 · +1 source units in pack.json

5 CLASS 40 Moment of Inertia_ Numerical

Study focus: CLASS 40 Moment of Inertia_ Numerical. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000042 · EV0000043

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

This is the first problem in today's class A slash 35.

Key details

- Determine the moment of inertia about the x-axis of the square area without and with the central circular pole.
- We have been asked to find moment of initiative this square about this x axis in the absence of this circular pole, and second.
- So the first part is in the absence of this, what is the moment of inertia of this square about the x-axis.
- If you look at this x axis, this x axis is.
- Nothing but centroidal x-axis because this square is symmetrical about this x-axis.
- That's why this itself will be the centroidal x-axis.
- Then how to find moment of inertia phase square about its centroidal x axis?
- We know we can make use of the formula of a rectangle itself.
- The moment of inertia of a rectangle about its centroidal x-axis is BH cubed by 12.
- four R and we can find IX.

- Square ear, x axis is here.
- This is without considering the central circular hole.
- What is B here?
- 4R and H is also 4R.
- So next I bar X will be equal to in this case it will be 6.
- 4 r to the power of 4 by 12.
- Simplify this, we are going to get it as 21.33 into r to the power of 4.
- This is the moment of inertia of this.
- Square about the x axis.
- About X axis that will be equal to moment of inertia of S square.
- About x axis minus moment of inertia of a circle above the same x axis.
- Pi r to the power of 4 by 4 because this itself is the centroidal x-axis even for circle.
- Cube by 12 minus pi into here radius is R itself.
- Sorry, R to the power of 4 by 4.
- This is the moment of inertia of the square with the central circular hole.

Example or application from the lecture

- They have given a square and a side 4R.
- We have to consider the circular hole here, and we have to find the moment of inertia of only the shaded area.
- Let's start with the solution.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000042 · EV0000043 · EV0000044 · +1 source units in pack.json

Core explanation · 05:00–09:59

Then this is the next question A slash 39.

Key details

- Calculate the polar radius of gallation of the shaded area about the center O of the larger circle.
- Look at this.
- There is a circle of radius 200 millimeter, and this is the center of the larger circle.
- Next, there is one more smaller circle.
- And C is the center of this smaller circle of radius, 100 mm.
- And if you look at these two O and C, they are on a same horizontal line.
- Here, they have asked us to find polar radius of gyration.
- Polar radius of gyration is nothing but Kz.
- Then how to find Kz?
- To find Kz, we have two formulas.
- One is under root of IZ by A.
- Where iz is polar moment of inertia and capital A is area.
- Then there's one more formula that is Kz is equals to under root of.
- Kx squar plus Ky squared.

- This formula we are going to use when we know the radius of variation about x axis and radius of variation about y-axis.
- Here, since we don't know the radius of variation about x-axis and y axis, we are going to use the first formula.
- That is Kz is equals to under root of IZ by A.
- Next, as I said before, iz is the polar moment of inertia.
- Since here we are talking about point O, that polar moment of inertia should be about the point O.
- Next, how to find polar moment of inertia of the shaded area about point about this point.
- Here, we want only the shaded area.
- Polar moment of inertia of this smaller circle about the point O.
- Then we are going to get the polar moment of inertia of the shaded area.
- Then find only the shaded area, then use the formula iz by A.
- We need to find Kz, that is polar radius of garation of the shaded area about 0.0.
- That is polar radius of garration about point.
- That is equals to under root of I z that is polar moment of inertia about the point.
- [ASR low confidence; check audio].
- Divided by total area capital E.
- Then how to find this polar moment of inertia that is Iz about point four.
- Polar moment of inertia about 0.4 will be equals to.
- I said about.
- O will be equals to polar moment of inertia of a bigger circle that is 200 mm radius circle about point O minus.
- Polar moment of inertia of a smaller circle that is 100 mm circle about the same point.
- Next, what is polar moment of inertia of this bigger circle?
- That is 200 mm circle, about 0.
- If you observe the figure carefully, that o itself is the center of the bigger circle.
- Polar moment of inertia about that point O will be equals to I bar X plus I bar Y.
- 4 plus I bar y is pi r to the power of 4 by 4.
- And what is radius?
- Radius is 200.

Example or application from the lecture

- Let's get into the solution.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000046 · EV0000047 · EV0000048 · +1 source units in pack.json

Core explanation · 10:00–14:59

200 to the power of 4 by 4 plus pi into 200 to the power of 4 by 4.

Key details

- Once we simplify this, we are going to get the value as 8 pi into 10 power 8 mm to the power of 4.

- This is the polar moment of inertia of a bigger circle about the point.
- So we are done with the first term.
- We need to find next.
- Polar moment of inertia of a smaller circle above the point O, that is Iz.
- I said of 100 mm circle about point O.
- This will be equals to here.
- Do we have a standard formula for polar moment of inertia about point O?
- No, we don't have standard formula, so we have to apply the parallel axis theorem.
- Remember, one of the axis is passing through O, that is Z-axis passing through O, and one more Z-axis is passing through C.
- The Z-axis passing through C is the centroidal z-axis of the smaller circle.
- Then what is i bar z?
- We know for a circle i bar z is pi r to the power of 4 by 4 plus pi r to the power of 4 by 4.
- Then what is distance?
- Distance is the distance between O and C, that is 15.
- Here, r is 100, so pi into 100 to the power of 4 by 4, plus pi into 100 to the power of 4 by 4, plus pi into 100 square into 50 square.
- Just use the calculator and find the value.
- We are going to get this as these two terms will be equal to 15 pi into 10 power 6.
- Then plus pi r square, that is pi into hundred square into fifty square, will get it as twenty-five pi.
- into 10 power 6.
- Then add these two, it will be 75 pi into 10 power 6 mm to the power of 4.
- This is the moment of inertia of a smaller circle about the point O.
- Next, I said of the shaded area.
- This will be equal to moment of inertia of the bigger circle, that is 8 pi into 10 power 8 minus moment of inertia of the smaller circle, that is 57 pi into 10 po 6.
- Seven, seven, five into ten per eight, mm to the power of four.
- Next, we should know the value of area.
- Here, area is area of the shaded region.
- That area will be equals to A is equals to.
- Pi r square minus pi r square.
- R square is 200 square minus 100 square.
- eight and a as thirty thousand into five and solve it we are going to get the value of K said as.

Example or application from the lecture

- Then plus area we have to consider here area is phi r square.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000050 · EV0000051 · EV0000052 · +1 source units in pack.json

Core explanation · 15:00–19:59

One five point four five millimeter.

Key details

- This is the radius of variation of the shaded area about the point O.
- That's it for the today's class.
- If you come across any queries, you can post your queries in PESO forum.

Example or application from the lecture

- Try to solve the problems given in the assignment as well as in the question bank.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000054 · EV0000055 · EV0000056 · +1 source units in pack.json

6 CLASS 41 Moment of Inertia_ Numerical

Study focus: CLASS 41 Moment of Inertia_ Numerical. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000058 · EV0000059

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

This is the question A slash 42.

Key details

- Calculate the moment of inertia of the shaded area about the x axis.
- We have been asked to find moment of inertia of only this shaded area about the x-axis.
- How to find inertia of this shaded area about the x-axis?
- What we can do here is we can find moment of inertia of this triangle about the x-axis, then in that we can detect the moment of inertia of this semicircle about the same x-axis.
- Then we are going to get moment of inertia of this remaining shaded area.
- About x axis.
- In this, we have to deduct minus moment of inertia of a semicircle.
- Of a semi circle about the same x axis.
- Next, what is moment of inertia effect triangle about x axis?
- Moment of inertia of a triangle about x axis, we know from basic formulas.
- That is equal to, I am going to call that as Ix1.
- That is moment of inertia of a triangle about x axis.
- That is equals to bh cube by.
- This is from the basic formulas we know.
- Here, what is B?
- B is the base of the triangle that is hundred.
- What is H?
- H is height of a triangle that is 90.

- Use the calculator, then find this value.
- That is six point zero seven five into ten per six.
- mm to the power of 4.
- This is the moment of inertia of a triangle about its base, that is about x-axis.
- Next we need to find the inertia of a semicircle about the same x-axis that I call it as Ix2.
- This will be equal to Since the X axis is passing through the.
- Is nothing but the diameter axis.
- We have a direct formula.
- Moment of inertia of a semicircle about its diameter axis is pi r to the power of 4 by 8.
- H r is 30.
- to the power of four by eight.
- Just the calculator find this value.
- This is zero point three one eight into ten power six mm to the power of four.
- This is the moment of inertia of a triangle about x axis minus moment of inertia of this semicircle about the x axis is 0.318 into 10 po 6.
- Once we solve this, we are going to get as 5.756 into 10 bar 6 mm to the power of 4.
- This is the moment of inertia of the shaded area about the x axis.
- This is the final answer.
- X question A slash 44.
- The variable h designates the arbitrary vertical location of the center of a circular cutout within the semicircular area.
- Determine the area moment of inertia about the x-axis for A H is equals to 0 and when H is equals to R by 2.
- In the first case, before that, look at this, there is a semicircle of radius capital R.
- And there is this circular cutout.
- When the position of the center of this circular cutout is h is equal to 0.
- That is when the center of the circle is coinciding with the origin O, in that case, what is the moment of inertia of the.

Example or application from the lecture

- Let's get into the solution.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000058 · EV0000059 · EV0000060 · +1 source units in pack.json

Core explanation · 05:00–09:59

Shaded area, nothing but what is the moment of inertia of the remaining area we have to calculate.

Key details

- Then in the second case, when H that is the position of the center of this circular cutout is R by 2, what is the moment of inertia of this?
- Remaining area about the x axis we have to calculate.
- when h is equal to 0.
- When h is equals to 0, how will be the figure?

- Since h is equal to 0, the half of the circle is going to be here.
- And the remaining half it's going to be outside this semicircle.
- The part which is outside the semicircle that doesn't matter to us because we are considering moment of inertia of the remaining area within this semicircle.
- The radius of this semicircle is capital R.
- Then the radius of this circle is R by 4.
- Then we are considering only this shaded.
- You need to find moment of inertia of only this shaded area about the x-axis.
- That will be equal to moment of inertia of this bigger semicircle minus moment of inertia of this smaller semicircle about the x-axis.
- Ix is nothing but moment of inertia of the shaded area about x-axis is equal to moment of inertia of a semicircle.
- About X axis.
- This is because the circle minus moment of inertia of a circle about the same X axis.
- What is this moment of inertia of a circle, semicircle about this x ax?
- We know moment of inertia about its diameter axis is pi r to the power of 4 by 8.
- N R is capital R minus.
- Moment of initial circle, since as I said before, half of the circle is outside this semicircular area.
- Because this x axis is his diameter axis even for the smaller semicircle.
- That's why.
- R is radius of the smaller circle that is R by 4 to the power of 4 by 8.
- Then deduct that from this, we are going to get the value of IS.
- 3911 into r to the power of 4.
- This is the moment of inertia of the shaded area about the x axis, and this is when h is equal to 0.
- This is the answer when H is equals to 0.
- when h is equal to r by 2.
- This is the semicircle of radius r.
- This is the center of the semicircle and this is the axis of symmetry y axis.

Example or application from the lecture

- Let's get into the solution.
- Let's start with the first case that is when h is equal to 0.
- Let's start with the semicircle.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000062 · EV0000063 · EV0000064 · +1 source units in pack.json

Core explanation · 10:00–14:59

This is the center of the circle.

Key details

- about the x axis.

- Then how to find that?
- In that we have to detect moment of inertia of this circle about the same x axis.
- Then only we are going to get moment of inertia of this shaded area about the x axis.
- Moment of inertia of a circ semicircle, moment of inertia of a semicircle.
- about X axis minus.
- Moment of inertia I'll write here itself moment of inertia of a circle about the same x axis.
- I'll call it this one as I x1 and I'll call this as I x2.
- Then how to find moment of inertia of a semicircle about the x axis that is i x1 that will be equal to.
- Since the x axis itself is the diameter axis for the semicircle, moment of inertia we have a direct formula that is I r to the power of 4 by 8.
- Yet what is R?
- R is R itself.
- This is the moment of inertia of a semicircle above the.
- This is the moment of inertia of a circle about this line.
- Since we don't have a standard formula, we have to apply the parallel axis theorem.
- Where will be the centroid of this circle?
- Centroidal x-axis of this circle.
- This is the center of the circle.
- Draw a horizontal line passing through this.
- This will be the centroidal x-axis of a circle.
- We are talking about moment of inertia of this circle about this axis.
- Inertia about any line is equals to inertia about parallel centroidal axis plus area into distance square.
- into distance square.
- What is I bar x per circle?
- I bar x per circle is pi r to the power of 4 by 4 plus area.
- That is area of a circle.
- That is pi r square.
- D stands for the distance between the two axes.
- Here one axis is X and one more axis is X0.
- That is R by 2.
- Just substitute here pi into r is nothing but the radius of the circle.
- What is the radius of the circle?
- Look at the figure, it is r by 4.
- Plus pi into r square is r is radius of the circle.
- That is r by 4.
- Square into distance is r itself.
- This is r.
- We are going to get this as five.
- 17 pi r pi into 17 r to the power of 4 divided by 1024.
- This is the moment of inertia of the circle about the x axis.
- pi into r to the power of four divided by one zero to four.
- We are going to get this as 0.341 into R to the power of 4.

- This is the moment of inertia of the remaining.

Example or application from the lecture

- For that, we have to consider moment of inertia of this semicircle about x axis.
- Let's start with this solution.
- It is given in the figure.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000066 · EV0000067 · EV0000068 · +1 source units in pack.json

Core explanation · 15:00–19:59

area about the centroidal, sorry, about the x axis which is passing through the diameter axis.

Key details

- This is the value ix is equal to 0.341 into r to the power of 4.
- That's it for today's class.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000070 · EV0000071 · EV0000072 · +1 source units in pack.json

7 CLASS 51.1 Moment of Inertia_ Numerical

Study focus: CLASS 51.1 Moment of Inertia_ Numerical. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000074 · EV0000075

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

Cushion A slash forty-four.

Key details

- Determine the moments of inertia of the Z section about its centroidal x axis and centroidal y axis.
- Look at this Z section.
- The width of this member is 100, and the depth is 40, the width of this web is 20, and the height will be from the bottom most member till here it is 140.
- Then there is one more rectangular section here.
- We have been asked to.
- Find the moment of inertia this Z section about the centroidal X and centroidal y axis.
- Then where will be the centroidal x axis?
- Due to symmetry, we can say that.

- The centroidal x axis will be at a distance of 140 plus 20 divided by 2.
- That will be equal to 80 mm with respect to the bottommost member.
- Then where will be this y naught axis?
- Y naught axis will be exactly at the center, again due to symmetry, that will be equal to from ear to ear it will be 100.
- From year to ear it is 100.
- R in one more way, hundred minus half of this width, that will be ten.
- Twenty by two will be ten.
- With respect to this member also, it will be at a distance of 90.
- Once we know the centroidal x axis and centroidal y axis.
- Next, we can find moment of inertia of this Z section about the centroidal x axis and centroidal y axis.
- If you draw a vertical line passing through this, this will be the centroidal y-axis.
- Then how to find moment of inertia of this satisfaction about this axis?
- We know the moment of inertia formula for rectangles.
- Either you can divide it either by drawing a horizontal line or by drawing a vertical line.
- This is the first rectangle, and here also I'm going to draw a vertical line.
- I'll call it as 2.
- Then this is the third rectangle.
- Now moment of inertia of this z section about centroidal x axis will be equal to moment of inertia of the first rectangle about centroidal x axis plus.
- Moment of inertia of the second rectangle about the centroidal x-axis plus moment of inertia of the third rectangle about the centroidal axis.
- I am going to start with the first rectangle.
- The first rectangle is like this.
- Here, what is this width of the rectangle?
- It is not 100 because this part is included in the second rectangle.
- This is B.
- Into what is the depth of the rectangle?
- The centroidal x axis of the z section is here, x0.
- And it is at a distance of.
- This is if you we have considered this is the x-axis.
- With respect to x axis, this is at a distance of.

Example or application from the lecture

- It's given this way like.
- Let's get into the solution part.
- As I said before, the coordinates of the centroid are if you consider this bottommost member as x-axis and the leftmost member as y-axis, then the coordinates of this centroid will be 90,, 80.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000074 · EV0000075 · EV0000076 · +1 source units in pack.json

Core explanation · 05:00–09:59

We know y bar is 80, so 80 at a distance of 80, it is there.

Key details

- Then, how to find inertia of this rectangle about the centroidal x-axis of the z section?
- We don't have a standard formula, so we have to apply the parallel axis theorem.
- To apply the parallaxis theorem, we have to.
- Look at the centroid of this rectangle.
- The center of the rectangle C is here, then what will be this distance?
- This will be 10.
- That is y bar is equal to h by 2, h is 20, 20 by 2, we will get as 10.
- Then from ear to ear distance we know that is 140.
- Whereas this is the centroidal axis of the z section.
- With respect to this rectangle, this is any other axis.
- What is I bar x formula for rectangle?
- That is bh cube by 12 plus area is B into H into distances, the distance between the centroidal axis of the rectangle and the centroidal x axis of the Z section.
- This distance we want that will be equal to 140 minus 80, we will get this distance.
- 140 minus 80, then plus 10.
- I'll write it here Bs eight into H s twenty B H cube by twelve.
- Area is B into H, that is 80 into 20 into distance square.
- Distance as I said before, that is 140 minus 80.
- That is 60, 60 plus 10, 70 square.
- Once we solve this, we are going to get the value as 7, 8, 9, 3, 3, 3, 3 mm to the board of 4.
- This is the moment of inertia of the first rectangle about the central x-axis of the z section.
- And one more, if you look at this.
- The second rectangle forget about the second rectangle for time being.
- If you look at the third rectangle, the centroid of the third rectangle will be here, and the distance from here to here will be 10.
- 10 from year to year it is 80.
- 80 minus 10 will give us 70.
- The centroidal x-axis of the z section will be equal to Ix.
- One moment of inertia the first rectangle about the centroidal z centroidal axis of the z section.
- Then we are left with only this rectangle.
- That will be equal to 140 plus 20.
- That is the height of this rectangle.
- That will be equals to 160.
- Both the centroid are coinciding with each other.
- The centroidal x axis of the z section and the centroidal axis of this second rectangle are coinciding.
- We know the standard formula that is BH cube by 12.
- Because this is 20 and h here is 140 plus 20.
- That is 160 cube divided by 12.

- And we'll get the answer as six eight two six mm to the power of four.
- Now we got the moment of inertia of the first rectangle, second rectangle and third rectangle about the centroidal x axis of the z section.
- So some just sum it up like Ix naught is equal to first piece Ix1 plus Ix2.

Example or application from the lecture

- Plus area we have to consider.
- If you consider this rectangle, where will be the centroid of the rectangle?

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000078 · EV0000079 · EV0000080 · +1 source units in pack.json

Core explanation · 10:00–14:59

Plus IX 3.

Key details

- If we add 1, then 2, plus again this, we are going to get the value of momentum inertia of the z section about the central x axis, that is 2261.
- 333 2 into mm to the power of 4.
- This is the moment of inertia of the z section about the centroidal x axis.
- Next, we need to find the moment of inertia of the same z section about its central y-axis.
- Here also we are going to apply the same concept.
- Moment of inertia of the Z section about its centroidal y axis will be equal to sum of the moment of inertia of these three rectangles about the same centroidal y axis.
- So first I'm going to start with the first so I.
- I y0 will be equal to I1.
- This is moment of inertia of the first rectangle plus I2, moment of inertia of the second rectangle plus I 3 moment of inertia of the third rectangle.
- So this is the rectangle, first rectangle, it's like this.
- Then here the width of the rectangle is 100.
- Then the height of the rectangle is 20.
- Next, where is the centroidal y-axis?
- The centroidal y axis is here.
- This is the y axis we have assumed in the figure.
- This is y axis, and this is the central y axis.
- Then the distance between these two is.
- Sorry, this is not hundred because this part is included in the second rectangle.
- And this is ninety.
- Because from year to year it is 80.
- 80 plus 10 is 90.
- That is what X bar value.
- Next, how to find inertia?
- This rectangle about this axis.
- Since we don't have a standard formula, we have to apply the parallel axis theorem.

- Where will be the centroidal y-axis of this particular rectangle?
- At a distance of x bar.
- X bar will be equal to b by 2 that is 80 by 2 will be 40.
- Inertia about any line will be equal to inertia about parallel centroidal axis plus area into distance square.
- This is the statement.
- Next, what is I bar Y?
- I bar Y formula for rectangle we know HB cube by twelve.
- Plus area is B into H.
- Distance is the distance between this centroidal y-axis of the rectangle and the y-nut axis.
- That will be equal to total 90, 90 minus 40.
- That will give us the distance between these two axes.
- Plus B is 20 and H is 80 into distances 90 minus 40, that is 50.
- We'll get this as four, eight, five.
- 3333 point three mm to the power of four.
- This is the moment of inertia of the first rectangle about the centroid and y-axis of the z section.
- Okay, now again here also we can apply the concept of symmetry because this centroidal y-axis is exactly at the same distance with respect to rectangle 1 and 3, and these two are identical, 1 and 3.

Example or application from the lecture

- Let's start with the first rectangle.
- Since the axis is vertical here, we should consider the centroidal y-axis of this particular.
- Let's start with the statement of parallel theorem.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000082 · EV0000083 · EV0000084 · +1 source units in pack.json

Core explanation · 15:00–19:59

Next, we are left with the second rectangle.

Key details

- Again, it's the same case now also.
- The centroid of this Z section itself is the centroid of this rectangle.
- That's the reason the centroidal y-axis of the Z section itself is the centroidal y-axis of the rectangle.
- What is h here?
- H is height of the rectangle that is 140 plus 20, that is 160 into.
- B's width of the rectangle 20 whole cube by 12.
- If we simplify we are going to get it as 10.
- 66.67 mm to the power of 4.
- This is the moment of initiative the second rectangle about the centroid and y axis.

- That is equals to moment of inertia of the first rectangle plus moment of inertia of the second rectangle plus moment of inertia of the third rectangle about the same central y axis.
- That is equals to it is 981.
- 333 3 into mm to the power of 4.
- This is the moment of inertia of the z section about the central y axis.
- If you come across any doubts, you can post your queries in the PESO forum.

Example or application from the lecture

- Consider that formula.
- try to solve the problems given in the assignment.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000086 · EV0000087 · EV0000088 · +1 source units in pack.json

8 CLASS 51.2 Moment of Inertia_ Numerical

Study focus: CLASS 51.2 Moment of Inertia_ Numerical. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000090 · EV0000091

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

This is question A slash 60.

Key details

- Calculate the polar radius of garage of the shaded area about its centroid C.
- If you look at this figure,.
- They have asked us to find polar radius of gradation, that is case about its centroid C.
- We don't know the coordinates of the centra, so the first step is we need to locate the coordinates of the centra here in this case.
- Here first, I am going to divide this into elemental areas.
- This is just for my reference, and the leftmost member is.
- Now the entire figure is in the first coordinate.
- First, I need to find the coordinates of the centroid C.
- Since this is Y axis, this is going to be X bar.
- And this is will be y bar.
- First, do we have the axis of symmetry here?
- Yes, vertical line.
- If I draw a vertical line here passing through this, this right of the section is the mirror image of left of the section.
- Total is 500.
- Bar is equal to 500 by 2, that will be 250 mm.

- Next, we need to know the value of y bar.
- To find y bar, we have a formula.
- Y bar is equals to summation of a i yi by total area.
- To apply this formula, we have to divide this into elemental areas.
- Then how to divide this into elemental areas?
- I am going to draw a horizontal line here and here also.
- Let it be first area.
- Let it be second area and let it be third area.
- We know here the formula is going to be a1 by 1 plus a2 by 2 plus a3 y 3 divided by total area.
- A1 is equals to A1 stands for area of the first rectangle.
- Base into height, since it is a rectangle, B into H.
- B we know 100 into H.
- What is H?
- H is from ear to ear distance.
- Since total is 600, total is 600.
- 600 minus this is 100, even in the figure.
- This will be fine to 10 power 4 mm to the power of mm square.
- After this we should do the value of y 1.
- This cent this rectangle will have it, this rectangle will have a centroid somewhere here that is C1.
- The y coordinate of this C1 will be equal to 100 plus H by 2 because from here to the base of the rectangle is.
- H by two.
- What is h here?
- H is five hundred.
- Next is a two.
- A2 is area of the second rectangle.
- This is the second rectangle.
- Again it is 5 into 10 power 4 mm square.
- Y2 is this rectangle will have a centroid C2 here, then this y coordinate of this is y2.
- Next, due to.
- These two are identical, like the same dimensions, they are at the same distance with respect to x.
- A3 will be equal to a1 and y3 will be equal to y1.
- If you want to calculate, you'll get the same value.
- Then substitute in y bar.

Example or application from the lecture

- I am going to consider this bottommost member as the x-axis.
- To find y bar, let's start with the first one.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000090 · EV0000091 · EV0000092 · +1 source units in pack.json

Core explanation · 05:00–09:59

A one is phi into ten power four into y one is three fifteen plus a two is phi into ten pour into y two is fifteen plus.

Key details

- A3 is A1 and Y3 is Y1.
- 2 times set of this divided by total area is A1 plus A2 again plus A1.
- Sorry, y we will get it as two fifty mm.
- This is the y coordinate of the centroid.
- Next we need to find moment of inertia of this.
- This is the centroidal x-axis.
- It is at a distance of y bar that is 250 with respect to bottommost member.
- How to find inertia of this C section about this centroidal x axis, which is at a distance of two fifty with respect to bottom most member.
- What we can do here is since we don't have standard formula, moment of inertia of this lamina about its centroidal x axis will be equal to.
- So, sum of the moment of inertia of the first rectangle, second rectangle, and the third rectangle about the same centroidal x-axis of this lamina.
- Will be equals to inertia of the first one plus second one plus third one about the same x0 axis.
- About snap axis.
- We know moment of initial about any line will be equals to moment of initial parallel central dial axis plus area into distance square.
- For time being forget about this second rectangle and the third rectangle.
- It's like this.
- The x-axis is here.
- Where is the centroidal x-axis of this rectangle?
- We know that is at a distance of y1, that is 350, because from here to 100 plus h by two will be 250.
- So that will be equal to 350 means above this line somewhere here.
- This is the centroidal x axis of this rectangle.
- Please remember when I'm talking about this rectangle, this is any other axis, whereas this will be the centroidal x axis.
- How to find initial any axis?
- Inertia about any axis will be equal to inertia about paral centroidal axis plus area into distance square and this is three fifty.
- Distance is 250 minus, sorry, 350 minus 250 is the distance between these two palms that will be equal to 100 one square.
- And substitute your B's width of the rectangle is 100.
- H is 500.
- cube byte 1 plus area is area of the first rectangle already we know 5 into 10 power 4 into distances 100 square.

- Once we solve this we are going to get the value of this and this is equal to 15.42 into 10 power 8 mm to the power of 4.
- This is the moment of inertia of only the first rectangle about the centroidal x axis of this entire lamina.
- Moment of inertia rectangle two about the centridal x axis.
- This is the centroidal x axis of the lamina x naught, which is at a distance of.

Example or application from the lecture

- given figure about its centroidal x-axis.
- Just consider this rectangle.
- Next we need to consider.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000094 · EV0000095 · EV0000096 · +1 source units in pack.json

Core explanation · 10:00–14:59

250 with respect to the bottom most member and the rectangle is here, rectangle 2 is here, and the dimensions of this are this is 500 from the figure, and this is 100.

Key details

- When we are considering this rectangle, this is any other axis with respect to this rectangle, and the central axis of this rectangle will be here, which is at a distance of 50 mm, because 100 by 2 will be 50.
- The central x axis will be equals to I bar x same formula I bar x plus area into distance squad.
- And distance is distance between the two parallelines, this line and the centroid and x-axis of the lamina.
- That will be equal to two fifty minus fifty.
- Two hundred whole square.
- Then here B is five hundred and h is hundred cube by twelve plus b into h b is five hundred and h is hundred into 200 squared.
- 42 into 10 power 8 mm to the power of 4.
- This is the moment of inertia of this rectangle, second one about centroidal x-axis.
- Again, next moment of inertia of the third one will be equal to moment of inertia of the.
- First one because these two are identical and they are at the same distance with respect to its axis.
- So I can write this will be equals to Ix naught will be equal to first one is Ix naught is equal to better I'll write it here I x naught is equals to.
- Moment of initial of the first rectangle is 15.42 into 10 power 8 and take it as common.
- Plus moment of initial of the second rectangle is 20 point four two.
- Moment definition of the third rectangle is again fifteen point four two into ten power eight.
- Moment of finish of the entire lamina about its centroidal x axis is fifth fifty one point two six into ten per eight mm to the power of four.
- This is the moment of inertia of the C section about the centroid and X-axis.

- Next, we need to find moment of inertia of the same C section about your centroidal y-axis.
- Again, same concept, moment of inertia of this lamina about its centroidal y axis will be equal to sum of the moment of inertia of the first rectangle, second rectangle, and the third rectangle about the same y-axis.
- The rectangle one is like this.
- Next, the centroidal y-axis is here.
- The centroid of the first rectangle is here, which is at a distance of 100 by 2 will be 15.
- From here to here it will be 15.
- Next, from this line to the central y axis of the lamina is 250.
- And this is the centroidal y-axis of this particular rectangle.
- And this will be any other axis with respect to this rectangle.
- Then inertia about any line will be equal to inertia about parallel centroidal axis, that is I bar y plus area into distance square.
- That will be equal to this we know 250.

Example or application from the lecture

- So let's start with first one.
- Moment of initial the rectangle one about its centroidal y axis of the lamina rectangle one I am going to consider.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000098 · EV0000099 · EV0000100 · +1 source units in pack.json

Core explanation · 15:00–19:59

So here h is the height of the first rectangle that is 500.

Key details

- Once we simplify this, we are going to get inertia of the first rectangle about its central y-axis of the lamina, that is equal to 20.42 into 10 power 8 mm to the power of 4.
- This is the moment of inertia of this section about the centroid.
- Sorry, moment of inertia of the first rectangle about the centroidal y-axis of the lamina.
- Since the centrate of this rectangle and the axis are coinciding with each other, no need to apply parallel theorem.
- Inertia of the second rectangle about this will be equals to this is directly it will be equal to HPQ by 12.
- No need to apply parallaxis theorem since we have a standard formula.
- Here H is 100 and B here is 500 Q by 12.
- Once we solve it, we are going to get the value as 10 point four two into ten power eight.
- This is inertia of a second rectangle about the centroidal y-axis.
- Next, again, these two are identical.
- So inertia of the third one will be equal to inertia of the first one.
- Moment of initiative the first rectangle is 20 point four two into ten per eight and that has common plus second one is ten point four two plus third one is again the first one twenty point four two into n power eight mm to the power of four.

- This is moment of inertia about the centroidal y axis.
- But the question is K Z radius of guiration about the centroidal y.
- About the point C, C is the centroid here.
- Plus I naught.
- That will be equals to.
- We know from the previous thing, I x naught is equal to that is I x naught is 51 point two six plus here.
- I y naught is fifty-one point two six.
- 26 into 10 by 8 is I z that will be equal to 102 point five two into ten bar eight mm to the power of four.
- This is I z.
- Next area we know area is a one plus a2 plus a three that is one five 000.
- Five two into ten part eight divided by one point five into ten part five.
- Once we solve this, we are going to get the values 261.43 millimeter.
- This is the polar radius of gradation passing to the center C.
- That's it for today's class.
- If you come across any queries, you can post your queries in the PESO forum.

Example or application from the lecture

- Next we should consider moment of inertia of the second rectangle about the centroidal y-axis.
- Just consider the rectangle 3.
- Try to solve the problems given in the assignment and question bank.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000102 · EV0000103 · EV0000104 · +1 source units in pack.json

9 CLASS 51.2 Moment of Inertia_ Numerical

Study focus: CLASS 51.2 Moment of Inertia_ Numerical. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000106 · EV0000107

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

So the first is moment of inner ship a rectangle about its centroid and x-axis.

Key details

- Let the height of this rectangle be h.
- Like the total area of this rectangle that is capital A will be equal to B into H.
- Next, we want to find inertia about the centroidal x-axis.
- First, we should mark the centroidal x-axis here.

- Where will be the centroidal x-axis?
- Then where will be the centroidal x axis?
- We know why by formula for rectangle is h by 2.
- The shaded area is the element.
- Since this is a small area, let it be dA.
- The distance is y.
- I x0 will be equal to we know moment of inertia's distance square into area.
- Here the distance is y, so y square into area is d.
- Next, this is the inertia of the smaller element about the centroidal x-axis.
- Next, I need to find the inertia of the rectangle about the centroidal x-axis.
- Once I integrate this, integral of d i x naught will be equal to i x naught.
- This is the moment of inertia of a rectangle above the centroid x axis.
- This will be equal to integral of y square into d.
- Integral of y square.
- into d a.
- What is d?
- D is the area of the small element.
- The area of the element will be equal to since it is in the form of rectangle, it will be equal to b into h.
- Here what is b?
- Since these two lines are parallel, the width of the element will also be equal to.
- B itself, so d a is equals to b into h.
- What is this h?
- The width of the element is a variable along y direction.
- Next, i x naught will be equal to integral of b's constant.
- Limits will be from C minus H by 22 plus H by 2.
- I x0 will be equal to B by 3.
- It will be equal to h by 2 4 cube minus of h by to cube.
- I x naught will be equal to b h cube by.
- This is the moment of inertia of a rectangular lamina above the centroid and x axis.
- That will be equal to bh cube by 12.
- You have to remember this.
- Moment of inertia of a rectangle about its centroidal x axis is b by 12.
- There is a rectangle like this, and here is the x-axis, and centroidal x axis is at a distance of.
- H by 2 from here.
- Moment of inertia about this axis, we got it as BH U by 12.

Example or application from the lecture

- For this I am going to consider a rectangle.
- Let us consider this rectangle of width B.
- If you consider this is just an x-axis.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000106 · EV0000107 · EV0000108 · +1 source units in pack.json

Core explanation · 05:00–09:59

We can apply parallel axis theorem.

Key details

- According to parallel axis theorem, inertia about any axis that is ix will be equals to inertia about parallel centroidal axis that is i bar x plus area into distance squared.
- This is the statement of parallel axis theorem.
- I bar X already we know B H cube by 12.
- Area is area of a rectangle that is B into H.
- Then D stands for the distance between the two parallel lines that is H by 2.
- Plus area is b into h, then the distance is h by 2 square.
- Just simplify this, we are going to get ix as this will be bh cube by 4, and this is bh cube by 12.
- [ASR low confidence; check audio].
- B.H.Q.
- B.H.Q.B.
- This is the moment of inertia of a rectangle above the centroidal, sorry, above the line passing through its base.
- Its bottom most member will be BHQ by three.
- Next is moment of inertia of a circle about its diameter axis.
- [ASR low confidence; check audio].
- Of radius R.
- Let O be the center of the circle.
- Let the radius of the circle let it be R.
- Because for circle, this is where the centroid.
- R0 with respect to center.
- Then the radial width of this element let it be DR0.
- Next, let the inclination of this element with respect to this centroidal x axis adjust the x-axis.
- Let it be theta, then the inscribed angle of this element let it be d theta.
- This is R0 and this is D R naught and this is D theta.
- Since it is like an arc and the arc this arc length will be equal to r naught into inscribed angle d theta.
- The area of this element it's on when D theta is small it means almost like a rectangle.
- This is the area of the smaller element.
- This is the inertia of the smaller element.
- Once we integrate this, we are going to get it as i x naught will be equal to integral of y square into d e.
- Next what is 1?
- D is R0 into DR0 into D theta.
- There are two variables.
- One, this element can be anywhere from 0 to R within this radius.
- At the same time, theta, this element can be anywhere from 0 to 2 pi.

- Y is nothing but this is the x-axis, it's like this, and this is y.
- When this d theta is very small, the element will be just like a point, then y will be the opposite side of this right angle triangle.
- This is theta, we know.

Example or application from the lecture

- For this I am going to consider a circle.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000110 · EV0000111 · EV0000112 · +1 source units in pack.json

Core explanation · 10:00–14:59

Zero two pine zero two or.

Key details

- Then R0 square into R0 will be equal to R0 Q into D R naught.
- This is the term related to R0, then sin square theta into D theta.
- Integration of R0 cube will be equal to R0 to the power of 4 by 4, and if you apply the limits from 0 to R, that term is going to be.
- R to the power of 4 by 4, one more term is anyhow 0.
- Then we need to integrate 0 to 2 pi sine square theta into d theta.
- Sin square theta will be equal to 1 minus cos 2 theta by 2.
- And for this the limits will be from 0 to 2 pi.
- Now this is IX0 is equal to R0 to the power r to the power of 4 by 4, this is first term.
- Then integration of 1 is.
- I'll take this 1 by 2 outside, so it's going to be 1 by 2.
- Then integration of 1 is theta because it is with respect to into d theta here and down d theta is there.
- Cos two theta will be equals to sine two theta by two, limits will be from zero to two by.
- Next, this will be equal to R to the power of four by eight.
- substitute it as two pi, two pi, then minus of sine of two times it of two pi divided by two minus of zero and zero.
- This will be equal to 5 R to the power of 4 by 8.
- This is the moment of inertia fit circle about the diameter axis.
- I x naught will be equal to i y naught.
- I can write it as i bar x is equals to i bar y is equals to pi r to the power of 4 by.
- See this is the moment of inertia of a circle about the diameter x-axis as well as the y axis.
- This is about circles.
- Only difference is the limits will be varying from zero to pi instead of two pi.
- here you can see the formulas for the semicircle.
- If you just derive it just like a circle by changing the limits from 0 to pi, you are going to get the values.
- Here, moment of inertia of a circle, sorry, semicircle, which is symmetrical about y-axis here.
- try to learn the formulas this way.

- Inertia about an axis of symmetry and the diameter axis is always equals, that is pi r to the power of 4 by 8 and pi r to the power of 4 by 8.
- Here, this is the centroid which is at a distance of 4 r by 3 pi.
- If you draw a horizontal line, this is the centroidal axis which is parallel to diametral axis.
- Moment of inertia about this line is 0.11 into R to the paraph.
- In that case, inertia about diameter and axis of symmetry is pi r to the power of 4 by 8, and the centroidal axis parallel to the diameter line is 0.11 into r to the power of 4.
- That's about semicircle.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000114 · EV0000115 · EV0000116 · +1 source units in pack.json

Core explanation · 15:00–19:59

In the derivation of circle, we are going to get ix and iy value.

Key details

- One is pi r to the power of 4 by 16, and this is also pi r to the power of 4 by 16.
- This is moment of inertia of a quarter circle about the radial lines.
- Next moment of inertia of a quarter circle.
- About the centroidal x axis and centroidal y axis is point five fi into r to the power of four.
- That's it for today's class.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000118 · EV0000119 · EV0000120 · +1 source units in pack.json

10 CLASS 47 Friction_ Numerical

Study focus: CLASS 47 Friction_ Numerical. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000122 · EV0000123

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

The first question for today's session, question number six slash eight.

Key details

- determine the coefficient of kinetic friction that is which will allow the homogeneous body to move down the thirty degree incline at constant speed.
- There is an inclined plane.
- Inclination of the plane with the horizontal is thirty degree.
- And this is a homogeneous body and we have to assume here it is.
- A two dimensional body.

- One end of this body is resting on a roller and the other end is resting directly on this inclined plane.
- Since this homogeneous body is resting on an inclined plane, here we are going to assume X axis along the inclined plane.
- We know y axis will be normal to x axis.
- Next, since we are considering this homogeneous body.
- It will be having a weight and we know weight will be always vertically downward.
- This W is an inclined force with respect to XY plane.
- Either with respect to x axis or with respect to y axis.
- Since this is thirty degree here, the angle between w and y axis will also be thirty degree.
- will be W sin 30 degree and the y component will be downward and magnitude will be W cos 30 degree.
- As I said before, this is a roller.
- Since the roller is.
- On an inclined plane we are going to neglect the frictional force.
- Normal to x axis and I'm going to call it as NA.
- That is the normal reaction at A.
- Next comes this part.
- It's like a foot here which is in contact with the incline.
- It will be having both frictional force as well as the normal reaction.
- I'm going to call it as N B.
- That is normal reaction at B.
- And because of this incline this.
- Homogeneous body will have a tendency to slide down the plane.
- To avoid that the frictional force will be up the incline.
- This is the frictional force at B.
- We can use the formula mu k is equals to F B by N B.
- So we need to find first what is frictional force at P and what is normal reaction at B.
- To find these two we have dot conditions of equilibrium.
- So first one is this f b is towards the positive side of x axis, so plus f b.
- This W sine thirty since it is in the opposite direction, it is minus W sine thirty.
- These are the only two forces which are along the X direction.
- C we will get the value of F B.

Example or application from the lecture

- And the figure is given like this.
- So first thing is we have to consider here X and Y axis.
- Next we need to consider y axis.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000122 · EV0000123 · EV0000124 · +1 source units in pack.json

Core explanation · 05:00–09:59

Point five times that we can write it as mg and we know sign that is point five so FP will be equals to point five times set of mg.

Key details

- Call it as equation one.
- Since we need to find mu k we need to.
- Know the value of F B and N B.
- It's not required to know the value of N A.
- Sum of the moments of all the forces about A as 0.
- That is the next condition of equilibrium.
- Moment due to F B will also be zero because it is a.
- the point is in the same line of action.
- And N B is acting in this direction.
- It is counterclockwise.
- Moment due to this normal reaction at B.
- That's what is written here.
- Next comes this W cos thirty.
- It is acting in the clockwise direction, so it should be taken as minus.
- Minus W cos thirty into perpendicular distance will be due to symmetry.
- This distance will be B by 2.
- Next comes this force.
- It is also clockwise.
- And the distance.
- We are talking about this distance.
- Again, due to symmetry, this distance will be B by 2.
- That should be equal to 0.
- And the equation is going to be N B will be equals to.
- Point six eight three times the dup M G and call it as equation two.
- We know next mu case.
- Yes, B by N B.
- This is the coefficient of kinetic friction between the homogeneous body and this incline for which this body will be under motion with the uniform speed.
- Question number six slash thirteen.
- The fifty kg wheels on its hub up the circular incline under the action of the twelve kg cylinder attached to a card around the rim.
- Determine the angle at which wheel comes to rest, assuming that friction is sufficient to prevent slippage.
- What is the minimum coefficient of static friction which will prevent this position to be reached with no slipping?
- here you can look at this figure.
- This is the fifty kg field, which is attached to a twelve kg cylinder.
- And due to this twelve kg cylinder, this fifty kg wheel rolls on this circular incline.
- Because here the friction between this circular wheel and this incline is sufficient for this to be in a state of rest.

- At the same time we need to find the coefficient of friction here between this 50 kg wheel and this circular incline.
- That's what we need to find.
- Since the weight of this circular wheel is 50 kg.

Example or application from the lecture

- Ah, which is the next condition of equilibrium we should consider.
- No need to consider sigma f y is equals to zero because we have two unknowns along y direction, one is n a and one more is n b.
- Distance here it is given as B.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000126 · EV0000127 · EV0000128 · +1 source units in pack.json

Core explanation · 10:00–14:59

Oh sorry, wheel is fifty KC that will be acting vertically downward.

Key details

- Fifteen to nine point eight one that is the weight of this wheel vertically downward.
- Next this wheel is in contact with here, so there will be a friction as well as the normal reaction.
- Since this will have a tendency to move this direction, the frictional force will be in the opposite direction, F will be up the incline.
- So next the same thing is considered here.
- there is a 12 kg cylinder attached to this wheel and the weight since it is freely suspended here the weight is going to be downward vertically downward to one into 9.81 N.
- I'll call this point as O and I'm going to apply the first condition as sum of the moment of all the forces about O as zero.
- This I'm going to start from here.
- It is acting in the clockwise direction with respect to this point of contact.
- So clockwise means negative, force is twelve into nine point eight one into distance.
- What is the distance between horizontal distance between this point of contact and the line of action of this force.
- That should be the horizontal distance, this distance.
- How to find this distance, if you look at this figure, the distance from the center of this wheel.
- Till here it is a radius that is two hundred millimeter.
- I am going to get that horizontal distance between the force and the moment center.
- This is the moment due to the 12 kg cylinder.
- Next moment due to F.
- And n will be zero because they are acting at the same point.
- self weight of the wheel since it is acting in the counterclockwise direction it should be taken as plus.

- This is W from here we know this is W into distance will be horizontal distance because the line of action is vertical the distance should be horizontal.
- I am going to call that distances.
- That should be equals to zero.
- Once I know the value of X1 here, then we will get the value of theta.
- Ah so how to find this X1?
- Is it possible to express X1 in the form of theta?
- Yes we can do that.
- Same AB V is taken here.
- If this angle is theta, then this angle will also be.
- Next A B as I said before that is X1.
- Then A U distance.
- A u is nothing but the radius of the smaller circle.
- The radius of the smaller circle we know is 75.
- sine theta, sine theta is equals to opposite by hypotenuse.
- X1 will be 75 sine theta.
- Substitute the value of X1 in the equation.
- Once we substitute the value of x1 in the equation one, we will get it as sin theta is equal to 0.516.
- Once we simplify this we are going to get the value of theta.
- Theta is thirty one point zero seven.
- Nothing but when the value of theta is thirty one point zero seven, the wheel which is rolling up this incline will be nest of.
- Rest because at this angle the frictional force here will be sufficient to prevent this from slipping under the motion.
- Next we need to find the value of F, coefficient of friction.
- For that we need to know the value of F and N.
- Once we know F.
- And n then mu will be is equal to f by n.

Example or application from the lecture

- Let's start with the first condition of equilibrium.
- I'm going to consider moment about this contacting point as zero.
- If you consider that the first one is.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000130 · EV0000131 · EV0000132 · +1 source units in pack.json

Core explanation · 15:00–19:59

Sigma f of x is equals to 0.

Key details

- Since this f is along this x axis itself, it should be taken as plus.
- F, so plus F, then W will be inclined force with respect to X and Y.

- And the inclination here will be since this angle is theta, angle with respect to Y axis will be theta since we are talking about x component of that's going to be w sine theta and the component is going to be left side so minus w sine theta.
- That should be equal to zero.
- Already we know the value of theta from the previous.
- Next, we should know the value of normal reaction.
- So we need to apply the second condition of equilibrium, sigma f of y is equal to 0.
- The y component of this W will be downward like this.
- Minus W into cos theta will be the component.
- Then Y component of this will be again downward.
- We know the value of theta, we need to find only the value of n.
- Once we know.
- Frictional force in the normal reaction, then coefficient of friction will be equal to F.
- We will get it as point six zero two.
- This is the answer.

Example or application from the lecture

- Here we are going to consider incline as like x axis will be along this incline like a tangent here.
- Then minus of we need to consider this, the x component of this will be again left side.
- When we consider sigma f y 0, n is upward, so plus n.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000134 · EV0000135 · EV0000136 · +1 source units in pack.json

11 CLASS 48 Friction_ Numerical

Study focus: CLASS 48 Friction_ Numerical. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000138 · EV0000139

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

Problem six slash eighteen The uniform slender bar as an ideal roller at its upper LDA.

Key details

- Determine the minimum value of the angle theta for which equilibrium is possible for mu s point two five and for mu s point five.
- We don't know the length of this bar, we can assume it as L.
- Let capital L be the length of this slender bar.
- And when muus is 0 to 5, then in that case this entire system will be in a state of equilibrium.

- Since it's a roller, there will be only one reaction from the surface onto this roller that will be normal to the surface.
- Next year the slender ball is resting on the surface, so I am going to assume it as a.
- Tough surface there will be two reaction normal reaction N B and one more friction.
- The direction of friction depends upon the tendency of motion.
- Because of this inclination the slender bar will have a tendency to slide this way.
- And we know frictional force will be as equals to mu s into n.
- Let it be L.
- Next, the weight of this center bar will be acting to the geometric center G that will be at a distance of L by 2 with respect to A, L by 2 with respect to B.
- Next, this is W vertically downward.
- W is vertical, F B is horizontal, N B is vertical.
- Whereas N A is inclined.
- For that I should know this angle here.
- Then how to find this angle?
- To find the angle I am going to drop it perpendicular to this point, then extend this horizontal surface here like this.
- Then you can see there is a right angle triangle A.
- C D this right angle travel this is 90 degree this is theta then if you look at this here this angle will be 90 so the remaining angle will be 15 degree if this is 15 degree this angle will be 15 degree.
- This is 15 degree.
- This will also be 15 degree and the component of this na will be right side na cos 15 and the component of NA along y direction will be upward and that will be na sine 15 degree.
- Before that, remember here right now here is point.
- Now the first condition of equilibrium is sigma f of x is equals to 0, considering right side component as positive and left side as negative.
- Start with this NE, NA is towards right side, so plus N A is N A cost 15, F B is left side, so minus F B is equals to 0.
- yes b can be written as 0.25 n b in that case n a will be equals to 0.25 n b1 cos 50 that will be equal to 0.2588 times that of n b I'll call it as equation one.
- Next, second condition of equilibrium sigma f of y is equals to 0 taking upward component positive, downward, negative.
- N component is upward, so plus n mu.
- Sine fifteen degree, then W is downwards so minus W, then N B is upwards so plus N B is equals to C.
- Next we know.
- N is 0.258 times N V from the equation one substitute that here into sine fifteen minus W plus N B is equal to zero.
- Then N B can be written as.

Example or application from the lecture

- This is the figure given.
- After this, let us consider the length of the slender bar.

- Going to consider like this.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000138 · EV0000139 · EV0000140 · +1 source units in pack.json

Core explanation · 05:00–09:59

N V will be equal to W divided by.

Key details

- One plus point two five eight into sine fifteen that will be equal to one point zero six seven zero.
- This will be equation.
- Next to the last condition of equilibrium is moment.
- Here I am going to apply moment about yes 0.
- Next comes the force W.
- W is acting in the clockwise direction, so it is minus W.
- This is force, W is force.
- Into perpendicular distance since the linear fraction of W is vertical, the distance should be horizontal.
- This distance if this is theta, this angle will also be theta, since this is L by two, A G distance is L by two, this will be equal to L by two cos.
- We are done with the moment of this W.
- Next comes F B, F B is acting in the clockwise direction.
- into distance since the line of function of f b is horizontal the distance will be vertical distance from this rectangle triangle A C B this side will become L sin 15 degree that will be the perpendicular distance for this force F.
- N sin 15.
- Next we are left with one more force NB.
- N B is acting in the counterclockwise direction.
- Since the line of action of N B is vertical, the perpendicular distance will be horizontal, that will be this distance.
- This will be equals to N.
- We have N B in the second term, we have N B in the third term also.
- Write the letters as it is L by two cos theta, then minus 12 psi n b sine fifteen degree, then plus of n b cos theta is equal to zero.
- In all the three terms we have N V and L.
- sine theta will be equal to 0.4665 cos theta sine theta by cos theta will be equal to tan theta that will be equal to 1.866 then theta will be equal to 61 point.
- Next question problem six slash twenty three eighty two kg man pulls the forty five kg card up the incline at steady speed.
- Determine the minimum coefficient of static friction for which the man's shoes will not slip.
- Also determine the distance S required for equilibrium of the man's body.
- If you observe this figure, there is an inclined plane.

- The inclination of the plane with the horizontal we can find it out, assume this is theta, then theta will be equals to tan inverse of 3 by 10.
- The inclination of this handle with respect to a line parallel to the incline plane in 40 degree.

Example or application from the lecture

- Moment about which part we have to consider here.
- Once we consider moment about A as 0, considering counterclockwise positive and clockwise negative, moment because of these two will be zero.
- Let's cancel that.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000142 · EV0000143 · EV0000144 · +1 source units in pack.json

Core explanation · 10:00–14:59

N I F by N in general.

Key details

- Then what is F?
- F is frictional force and N is normal reaction.
- Then after that we have to find this yes.
- S is the horizontal distance between the A.
- That is the.
- Position of the man's shoe with respect to the center of gravity of man.
- This is also unknown.
- Once we know the force in this handle from the free body diagram of cart, the same force will be acting throughout from this point to the other point.
- normal to the incline plane.
- Next is weight of the cart.
- The self weight of the cart will be always acting vertically downward I'll call it as WC.
- WC stands for weight of the card that is equals to mass of the card 45 into.
- This will be m.
- Next, this is inclined.
- As I said before, this will be theta, so this will also be theta.
- Then theta is equals to tan inverse from the triangle we know 3 by 10.
- Next, after this there will be a force in this hang.
- That force is unknown right now, I will call it as U.
- Then the components of this will be along the positive direction of x axis that will be f cos 40 and one more will be in this way f sin 40 next by considering the free body diagram of cart.
- We want to find only the value of F we only need to find the reaction at these two wheels or rollers.
- This f is in the positive direction of x axis, so it will be plus f cos.
- Then this W Wc the component of this WC is down the incline, so minus WC, since the angle is with respect to y axis, it will become sine.

- As I said before.
- I need to find only the value of f so after simplifying this f will be equal to 165.6 newton.
- It is surface here, it will there will be having two reactions.
- One is normal reaction, one more is frictional force, one is along and one more is normal to the nuclear.
- Then at G the weight of the man will be acting that is vertically downward weight of the man w m will be equals to 82 that is mass of the man into acceleration due to gravity that is 9.81 newton.
- Next there will be a force through this member through the handle that will be equal and opposite to this F.
- But once we come to Wn, this is inclined with respect to y and x axis.
- To resolve this, since this angle is theta, this angle will also be theta this angle will be theta the components are going to be like this will be wm cos theta this is the component along y direction and one more will be w m sine theta.

Example or application from the lecture

- After that, once you consider a man, we don't know the force acting in this handle of the card.
- That's why let's start with the free body diagram of the cart.
- Once we consider a card let's start with x axis that is along the inclined plane and y axis.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000146 · EV0000147 · EV0000148 · +1 source units in pack.json

Core explanation · 15:00–19:59

This will be W M sine theta.

Key details

- The component is going to be f cos 40 and one more component will be log y direction f sine 40 d.
- Sigma f of x is equal to zero.
- This n is towards the positive side, so.
- F a is towards the positive side, so plus then f is in the opposite direction.
- then w is w component is also in the opposite direction so minus w sine theta is equals to zero so f then minus f is 165.6 into cos 40 then minus w m is 82 into 9.81 into sine theta is equal to 0.
- Then we know everything except f a so we can calculate f a f a is equal to 358 point nine Newton.
- Next we have to find.
- The one more force that is ne, apply the second condition of equilibrium, sigma f y is equals to 0.
- F into sin 40 is equal to 0.
- Here except NA we know all the other terms, so N A will be equal to 870 6.93 Newton.
- Once we know F A then N A we know mu.

- Coefficient of friction muis will be equal to all this, F A by N A.
- This will be equals to 0.409.
- We got the coefficient of static friction.
- Next we should know the value of yes.
- S we will get only when the moment equation comes because moment is forced into perpendicular.
- Here when we talk about momentum.
- M into perpendicular distances yes.
- here what do we know here is this angle with respect angle here will be theta.
- If this is theta and this is theta 40 this angle will also become theta plus 40 degree.
- Now apply the condition moment about D is equals to zero.
- Since this is in the clockwise direction, it will be minus force is E of sine of.
- Theta plus 14 into perpendicular distances 175.
- Then again it is clockwise so minus of cos.
- theta plus 40 into perpendicular distance will be equal to 850 that is equal to 0.
- Here we know the value of f we know the value of wm we know the value of theta so we just need to find the value of s so s will be equals to 126.2 meters.

Example or application from the lecture

- In the same vein, if this is 40 degrees, it's given here if I draw a line parallel to incline this will be 40 degree.
- So apply the next condition of equilibrium considering moment about any point zero, here we are going to consider moment about this point A as 0.
- we know need to consider these components along X and Y.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000150 · EV0000151 · EV0000152 · +1 source units in pack.json

Core explanation · 20:00–24:59

That's it for today's class.

Example or application from the lecture

- Try to solve the problems given in the cushion bank as well as the assignment.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000154 · EV0000155 · EV0000156 · +1 source units in pack.json

12 CLASS 56.1 Friction_ Numerical

Study focus: CLASS 56.1 Friction_ Numerical. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000158 · EV0000159

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

Question 6/33 Determine the distance S to which the 90 kg painter can climb without causing the 4 meter ladder to slip at its lower end A.

Key details

- The top of the 15 kg ladder has a small roller.
- And at the ground, the coefficient of static friction is 0.25, the mass center of the painter is directly above our feet.
- This is the figure we were supposed to follow for this particular question.
- 25 that is mu s is equal to 0.25 at the ground.
- Then at the B, this roller is resting on the vertical wall and we are going to neglect the frictional force at this roller because we will assume it is an ideal condition.
- She is trying to climb the ladder.
- When she is at a distance of s with respect to A, then she will be in a state of equilibrium.
- If she further climb up the ladder, then the ladder is going to slip.
- In this direction.
- We know the length of the ladder is 4 meter.
- I call this point as G.
- G is the center of gravity.
- of the ladder through which the weight of the ladder will be acting vertically downwards.
- W stands for weight of the ladder.
- That will be equal to mass of the ladder into G.
- The reactions here at A, since the ladder is directly resting on the ground, there will be normal reaction NA as well as there will be a frictional force.
- Since the ladder is having a tendency to slip in the left side direction, frictional force here will be right side here, A.
- Next, we know the coefficient of friction at A, so F A can be written as mu into N A.
- Next and B there will be a normal reaction from the wall onto the roller.
- So NB will be horizontal because the wall is vertical.
- Weight of the painter will be equal to mass of the painter, that is 19 to 9.81 newton.
- Then after that we can find the value of S.
- The first condition of equilibrium here is sigma f of x is equals to 0 considering the right side component of the forces positive left side negative.
- F is towards right side so plus e of a then n b is towards left so minus n b is equal to 0.
- These two are the only forces along X direction.
- What is FA here is 0.25 times the of NA.
- That will be equals to N B.
- R in other ways N B is equals to 0.25 times the of.
- And I'll call this as equation one.
- Na is upwards so plus Na, then weight of the painter is acting downwards, so minus WP, weight of the ladder is acting downwards, so minus W L is equal to 0.
- Here we know WB that is 90 into 9.81.

- We know WL that is 15 into 9.81.
- After substituting WB and W L here we will get the value of NE.
- That is 1030.5 newton.

Example or application from the lecture

- Here what is given is there is a ladder AB, the length of the ladder is 4 meter, then at A the ladder is directly resting on the ground, that's why they have given the coefficient of static friction between the ladder and the ground as point.
- There is a painter, her mass is given as 90 kg.
- I am going to consider the ladder for the free body diagram.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000158 · EV0000159 · EV0000160 · +1 source units in pack.json

Core explanation · 05:00–09:59

We can substitute this value of NA in the equation 1 and we can find the value of N B.

Key details

- Next, the last condition of E helium is moment.
- Since at A we have two forces, I am going to apply moment at A as 0 so that I can apply those two forces in this equation.
- Moment of this W is.
- W P into distance.
- This distance I will call it as X1.
- Then how to find this distance X1?
- To find this distance X1 I should know this angle theta.
- Then how to find this angle theta?
- I'll correspond to C.
- Theta will be equals to I know the hypotenuse, I know the adjacent side, so it will be equal to cos inverse of adjacent is 1.5 by hypotenuse is 4.
- This is the value of theta.
- Moment because of this WP is clockwise, so minus WP.
- This is the force into perpendicular distances x1.
- And what is X1?
- X1 will be equal to S cos theta.
- From the small triangle here.
- Then come to this WL.
- W n is also acting in the clockwise direction.
- This distance is 2 and this angle is known to as theta, this will be equal to 2 cos theta.
- W n into 2 cos theta.
- This is the moment because of the weight of the ladder.
- Then it comes to this NB, it is acting in the counterclockwise direction.
- Since the line of action is horizontal, the distance will be vertical, that is this C D.
- Side C B.

- Then how to find the C B?
- We know C B will be equals to under root of 4 square minus 1 point five square because a C B is the right angle triangle.
- X1 we know S cos theta.
- Then minus weight of the ladder.
- Weight of the ladder is 15 into 9.81.
- That is equal to 1 for 14.
- one forty seven point one five into two cos theta is sixty seven point nine seven then plus N B N B we know N B is two fifty seven point 512 into C B.
- C B is under root of 4 square minus 1.5 square that is 3.708 will be equal to 0.
- In this except yes, we know all the other terms.
- If she further climbs up the ladder, then the ladder is going to slip down the plane.
- That is what the meaning.
- Next question 6 slash 36.
- The homogeneous square body is positioned as shown in the figure.
- If the coefficient of static friction at base power four determine the critical value of angle theta which will which slipping will below which slipping will occur.
- Then friction at A.

Example or application from the lecture

- To find the theta, I am going to consider this big triangle A, this is B.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000162 · EV0000163 · EV0000164 · +1 source units in pack.json

Core explanation · 10:00–14:59

Here what is given here is there is the homogeneous square body and that body is having a roller at A and the body is resting directly on this horizontal surface at B.

Key details

- The side A B of this body is making an angle of theta with respect to this horizontal surface.
- It is that value of theta for which the body will be in a state of equilibrium.
- If there is a if the value of this theta decreases, then the body is definitely going to slide or slip.
- Along this surface, it's going to move in this direction.
- Since the body is in direct contact with the surface, there will be a normal reaction as well as there will be a friction force.
- This body will have a tendency to slip in the right side direction, the frictional force here will be towards left side.
- I will call that as FB.
- FB will be equal to mu into normal reaction.4 N this year normal reaction is NB.
- Next, there will be.
- Reaction from this surface onto this roller that will be always making an angle of 90 degrees with the surface.

- Normal reaction NE.
- Normal reaction at A.
- For that I am going to first find out the geometric center of this.
- Then I am going to.
- Since it we don't know exactly what is this mass or weight, I'm going to call it this is just W.
- This is along its direction, so we no need to find the angle.
- This is along y direction, no need to find the angle.
- But here it is inclined.
- This is 60 degree.
- I am going to draw a horizontal line here.
- If this is 60, this angle will become 30 because the angle between this norm reaction and the surface is 90 degree.
- That's why this angle will become 30 degree.
- Then we will find the value of theta.
- X component of this Na will be towards right side.
- Then F B is towards left side.
- I can write here N B is equals to 2.165 times of NE.
- This is the relationship between N and we will call it as equation one.
- The upward component of N A will be Na sin thirty.
- Since it is upwards it's plus Na sin thirty and W is acting downwards so minus W then N B is acting upwards so plus N B is equals to 0.
- So N A will be equals to once N A will be equals to W divided by 2.664 because we know N B is 2.165 times zero in A.
- We will get this expression and call this is equation 2.
- Next, the last condition of equilibrium, moment about any point 0, I'm going to apply this at B because at B I have two forces.

Example or application from the lecture

- Let's start with the free body diagram of this homogeneous body.
- Next we have to consider the weight of this body, homogeneous body.
- Consider draw one line the distance of s by 2 then s by 2.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000166 · EV0000167 · EV0000168 · +1 source units in pack.json

Core explanation · 15:00–19:59

Two components, one is horizontal and one is vertical.

Key details

- This is NA cos thirty and this is NA sin 30.
- This is force.
- Into perpendicular distance we need to find.
- Then how to find the perpendicular distance.

- You can see there is a rectangle triangle here, it's like this.
- This side is this is A and this is B.
- A B we know it's the one of the side of a square, it will be S and this angle we know theta.
- Then if I call this point as C then this will be equal to S cos theta and this will be equals to S sine theta.
- That's why here for this particular force the perpendicular distance will be the height of this triangle that will be equal to s sin theta.
- S sine theta.
- Then come to this force it is acting in the clockwise direction again.
- Perpendicular distance will be the base of this channel that will be S cos theta.
- Next come to this W for W.
- You have to remember one thing.
- This distance that is also S by 2.
- Since we know the coordinates of this particular point G, that is the point to which.
- The mass of this body is acting vertically downward.
- Since this is S by 2 and S by 2, we know the coordinates of this G with respect to B.
- That's why resolve it along this line and normal to this line.
- For this, we should know the angle.
- The angle here will be connect.
- We have to look into this figure.
- This angle is theta.
- If this is theta, this angle will be also theta because mutually perpendicular.
- Cos theta this will be W sine theta.
- It is clockwise so minus W sine theta.
- perpendicular distance will be S by two.
- Then come to the component.
- This component it is counterclockwise, so plus W into cos theta into perpendicular distance will be s by 2.
- This will be equal to 0.
- This will be.
- We have sine theta here, we have sine theta here also.
- And we have cos theta here and we have cos theta here.
- Once we simplify this, we are going to get the expression like this.
- Sine t sine theta into point A to five will be equals to cos teta into point three one 23 then sin by cos will be equals to tan theta that will be equals to 0.3785 then theta will be equals to tan inverse of 0.3785 that is equals to twenty point seven three.
- This is the value of theta for which this homogeneous body will be in a state of test.
- If there is a decrease in the value of theta, then this is going to slide down the plane.
- That is what the meaning.
- That's it for today's class.

Example or application from the lecture

- Let's start with the horizontal component, it is acting in the clockwise direction.
- For that I am going to consider this right angle triangle.

- If you consider W as it is, then we don't know this perpendicular distance and we can't find it so easily.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000170 · EV0000171 · EV0000172 · +1 source units in pack.json

13 CLASS 56.2 Friction_ Numerical

Study focus: CLASS 56.2 Friction_ Numerical. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000174 · EV0000175

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

Problem six slash three.

Key details

- The force peace applied to the 50 kg block when it is at rest.
- Determine the magnitude and direction of the friction force exerted by the surface on the block if the.
- P is equals to zero, second P is equals to two hundred newton, third case when P is to fifty newton, fourth, when what value of P is required to.
- Initiate motion up the incline.
- The coefficient of static and the kinetic friction between the black and the inclined mu is equals to 0.25 and mu case equals to 0.2 respectively.
- Then there is a fifty kg crate resting on this inclined plane.
- It is in a state of rest.
- Then we are going to apply this force P.
- Due to this force, this block will have a tendency to move up the.
- Then in the last they have asked us for what value of this P this block will start to move up the inclined plane.
- That's when p is equals to zero.
- Next, the weight of this block will be vertically downward.
- Then the components of this W parallel x axis will be up the incline sored plane and the component of W.
- Parallel to y axis will be downward like this.
- Then P, the components of P will be right and upward.
- Since here it is P is 0, you can neglect.
- This for time being, then there will be a normal reaction from the surface onto the plane that will be upward.
- Since the value of P is 0 this block will have a tendency to slide down the plane.
- This is the complete free body diagram of the block.

- Since n is upward, it will it will be plus n and the y component of w is downwards, so minus w into sine so cos 15 degree because the angle 15 is with respect to.
- Why the axis so cost 15 degree.
- Now apply the second condition of equilibrium sigma if x is equal to 0.
- Then the x component of this w is down the plane, so that will be minus w into sine fifteen degree.
- From this we can find the value of F.
- F is 126.95 newton.
- That we will get to know.
- Only when we compare this frictional force with the maximum frictional value.
- We know f max is equals to mu s into n, here mu s is 0.25 and n we know from here it is 475.78.
- Simplify we are going to get the value of f maxes under an 18.44 newton.
- Since F is greater than F max, this condition is not at all possible.
- It is impossible.
- If the block is moving down the plane, then the friction between the block and the surface will be.

Example or application from the lecture

- Here what is given is there is an inclined plane.
- This inclination of the plane with the horizontal is given as fifteen degree.
- And they have given the coefficient of static friction as 0.25 and kinetic friction as point two.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000174 · EV0000175 · EV0000176 · +1 source units in pack.json

Core explanation · 05:00–09:59

Kinetic friction that will be equals to mu c and two n.

Key details

- Here mu k is point two and n we know four seventy three point seven eight.
- So actual friction between the block and the plane when p is equals to zero is seventy four point seven five newton and that is up the plane because the block is moving down the plane.
- Next, the second case when the value of P is two hundred newton.
- Like this, x axis and the y axis in this direction, normal to the x axis, then the cell fate vertically downward, then the components of W will be like this, then components of P going to be P cos 20 and P sin 20.
- Normal reaction up the plane and the frictional force will be.
- Since the value of P is 200 newton we are going to assume because of this P this block will have a tendency to move up the plane, so the frictional force will be down the plane.
- That's why I have considered the friction in this direction.
- Then again the same conditions of equilibrium, sigma f of y and sigma f of x is equal to 0.
- N is upwards to plus n.

- This P sin 20 is also upwards, so plus P sin 20, W is downward, so minus W into cos 15 degree will be equal to 0.
- Here only one unknown that is N.
- Substitute the value of P s 200, we'll get the value of NS 542.2 newton.
- Next sigma Fx is equal to 0.
- X component of W is towards left, so minus W into sin twelve fifteen will be equals to zero.
- Here also only one unknown that is F.
- Further what we can do here is we have to compare this.
- Value of F with respect to F max.
- We know that F max is equal to mu S into N.
- N muus is point two five and n is five forty two point two.
- Find the value of F max.
- We'll get F max as 135.55 newton.
- With respect to F max, since F is smaller than F max, this indicates the body is in a state of rest.
- Once the body is under rest, the frictional force will be this itself, that is.
- Sixty point nine N and since the answer is positive that indicates our assumption is correct.
- We assumed the block is having a tendency to move up the plane, so frictional forces down the plane.
- This is correct.
- The third condition.
- When P is two fifty N, the same wavy body diagram.
- Since the value of p is two fifty newton we will assume the block is moving up the plane so fictional force will be down the plane.
- Apply the conditions of equilibrium sigma f y 0.
- N is upward, so plus n, then p sin twenty is also upward, so plus p sin 20, then the y component of w is downwards, so minus w.
- Cause fifteen will be equal to zero.
- Here also only un one unknown that is n.
- Next second condition of equilibrium, sigma f of x is equal to zero.
- f is towards left, so minus f.
- P component is towards right.
- Here also only one unknown that is CF.
- We'll get the value of F as 107.97 N.
- That we will get to know only when we compare the value of this F with respect to F max.
- N is double five nine point three and s is point two five we'll get the value of f max is one thirty five point five newton.
- Since f is smaller than f max this indicates the body is under rest.
- So therefore f will be equals to.

Example or application from the lecture

- It is the same free body diagram normal to the consider x axis along the plane, but.
- Let's start with sigma f y 0.

- Once we consider sigma fx is equal to 0, f is left side, so minus f, x component of p is towards right, so plus p cos twenty.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000178 · EV0000179 · EV0000180 · +1 source units in pack.json

Core explanation · 10:00–14:59

107.95 newton and we assume the block is having a tendency to move up the plane so fictional forces down the plane.

Key details

- And we got the answer also as positive.
- This is the last case for what value of P this fifty kg block will start to move up the incline.
- That's what we have to calculate.
- The solution remains the same.
- Again we have to start with a free body diagram.
- Components of W are going to be like this.
- Then the components of P are going to be right side and upward, then up.
- It's going to be P cos twenty and this is going to be P sin twenty.
- Normal reaction will be upward.
- Since the block is moving up the plane, the frictional force will be down the plane.
- Next, start with the conditions of equilibrium.
- Sigma F of Y is equals to 0.
- N is upwards, so plus N.
- P component of P is upwards, so plus P sin 20, component of W is downwards, so minus W.
- Cos 15 will be equals to 0.
- Here we have two unknowns, one is P and one more is N.
- Call it as equation one.
- Now apply the second condition of equilibrium that is sigma fx is equal to 0.
- Since the component of F is left side, so minus F, it is right side, so plus P e c cost twenty, it is left side, so minus W sine fifteen will be equal to zero.
- Here also we have two unknowns F and P.
- Since the block is under motion, we can replace F as mu K into N because here the frictional force will be kinetic friction that will be always equal to mu K into N.
- Once we solve equation one and two, we are going to get the value of P s two thirty nine point three newton.
- That is for the meaning.
- Next question, problem 6 slash 6.
- here as I said before always x axis should be taken along the inclined plane.
- The components of W are going to be X component is going to be down the plane and the Y component is going to be downward and the angle here, since this angle is 15 degree, the angle between W and the Y-axis will also be 15 degree.
- Next there will be a normal reaction from the surface unto the drum that will be upward and before that.

- P is that so we have to resolve this P.
- To resolve the P we should know the angle.
- If I draw a line passing through this, which is parallel to X axis, the angle between the P and the line parallel to X axis will be fifteen degree because that is going to be thirty minus fifteen, that will be fifteen degree.
- The component is going to be P cos fifteen and one more will be P sine fifteen degree.
- The next is normal reaction of the plane and it is going to move in this direction, the drum is moving in this direction, so the frictional force will be up the incline plane.
- Next, we have to apply the conditions of equilibrium.
- If you think of sigma Fx and FY, we have two unknowns.
- P is also your unknown and n is also unknown.

Example or application from the lecture

- Consider X along the plane and the Y axis normally to the plane and the weight vertically downward.
- Here mu K is given as.
- And the weight since the mass is given as m, we have to consider the weight w vertically downward.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000182 · EV0000183 · EV0000184 · +1 source units in pack.json

Core explanation · 15:00–19:59

If you think of sigma fx zero, f is a unknown, but we can express f as mu s into n, that is correct.

Key details

- But N is also an unknown.
- W cos fifteen, then P sin 15 will be equal to 0 because they're acting at the same point.
- Since P is clockwise, it is going to be negative.
- Minus P cos 15 into perpendicular distance will be R by 2.
- Then moment of F will be equal to this counterclockwise direction.
- Once we apply sigma fx 0 plus f minus w sine 15 plus p cos 15 will be equals to 0.
- As we know the value of f substitute here f is point four eight three times that of p minus w is m into g into sin fifteen plus p cos fifteen.
- Solve this we are going to get the value of P as 0.78.1786 times it of mg.
- Next apply the next condition of equilibrium sigma f y 0 n is upwards so plus n then p sin 15 is also upward so plus.
- Then W cos 15 is downwards or minus w will be equal to 0.
- We already know the value of P and P in terms of Mg, so substitute here W is also equal to Mg.
- We will get the value of NS 0.919 times of Mg.
- Next we should know the value of coefficient of static friction mu s it will be equal to f by n.
- F is 0.483 times the of P and N is 0.919 times the of EMG.
- We'll get the value of mu s as 0.094.

- This is the coefficient of static friction between the drum and the surface for which this will have a tendency to roll up the plane.
- That's it for today's class.

Example or application from the lecture

- So let's start with the moment condition.
- We can consider moment about the center of the drum as 0.
- Once we consider moment about O as 0, the moment due to W sine 15, then.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000186 · EV0000187 · EV0000188 · +1 source units in pack.json

14 Source slide deck: Moment of Inertia

Study focus: Source slide deck: Moment of Inertia. The notes identify each page heading, summarize its main ideas, and explain its place in this unit; citations preserve the source page locator. EV0000210 · EV0000211

SHORT-NOTE RECAP

Study notes · Source slide deck: Moment of Inertia

The slide deck is rewritten here as searchable lesson notes. Each subsection states the slide heading and its concise teaching points; page citations lead back to the indexed source image when visual details are needed.

Slide 1 · ENGINEERING MECHANICS

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 2 · ENGINEERING MECHANICS

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 3 · Engineering Mechanics

- ●Rectangular moment of inertia/ second moment of inertia
- ●Radius of Gyration(Rectangular radius of gyration and polar radius
- ●Derivation of moment of inertia for different geometric figures

Slide 4 · Engineering Mechanics

- ●First area moment of inertia is either positive or negative or zero
- ●Second area moment of inertia is the Moment of first area moment of
- ●but Second area moment of inertia is always positive about the axis

Slide 5 · Engineering Mechanics

- Rectangular Area moment of inertia and polar moment of inertia

Slide 6 · Engineering Mechanics

- Rectangular Area moment of inertia and polar moment of inertia

Slide 7 · Engineering Mechanics

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 8 · Engineering Mechanics

- ●The moment of inertia of an area about a non centroidal axis may be
- easily expressed in terms of the moment of inertia about a parallel

Slide 9 · Engineering Mechanics

- ● By definition, the moment of inertia of the element dA about the x

Slide 10 · Engineering Mechanics

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 11 · Engineering Mechanics

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 12 · Engineering Mechanics

- ●Conditions for parallel axis theorem
- 1. two axis should be there and two axis must be parallel to each other
- 2. Between two axis, one axis has to pass through the centroidal axis

Slide 13 · Engineering Mechanics

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 14 · Engineering Mechanics

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 15 · Engineering Mechanics

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 16 · ENGINEERING MECHANICS

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 17 · ENGINEERING MECHANICS

- A/1 Determine the moments of inertia of the rectangular area about the x- and y

- axes and find the polar moment of inertia about point 0.

Slide 18 · ENGINEERING MECHANICS

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 19 · ENGINEERING MECHANICS

- A/3 The narrow rectangular strip has an area of 300 mm2, and its moment of inertia
- about the y-axis is 35(103) mm4. Obtain a close approximation to the polar radius of

Slide 20 · ENGINEERING MECHANICS

- A/4 Determine the ratio b/h such that Ix = Iy for the area of the isosceles triangle.

Slide 21 · ENGINEERING MECHANICS

- A/6 Determine the polar moments of inertia of the semicircular area
- about points A and B.

Slide 22 · ENGINEERING MECHANICS

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 23 · ENGINEERING MECHANICS

- A/7 Determine the moment of inertia of the quarter circular area about the y-axis.

Slide 24 · ENGINEERING MECHANICS

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 25 · ENGINEERING MECHANICS

- A/9 The moments of inertia of the area A about the parallel p- and p'-axes differ by
- 15(106) mm4. Compute the area A, which has its centroid at C.

Slide 26 · ENGINEERING MECHANICS

- A/35 Determine the moment of inertia about the x-axis of the square area without
- and with the central circular hole.

Slide 27 · ENGINEERING MECHANICS

- A/36 Determine the polar moment of inertia of the circular area without and with

Slide 28 · ENGINEERING MECHANICS

- A/37 Calculate the polar radius of gyration of the area of the angle section about point
- A Note that the width of the legs is small compared with the length of each leg.

Slide 29 · ENGINEERING MECHANICS

- A/38 Calculate the polar radius of gyration of the area of the angle section about point
- A Note that the width of the legs is small compared with the length of each leg.

Slide 30 · ENGINEERING MECHANICS

- A/39 Calculate the polar radius of gyration of the shaded area about the center 0 of

Slide 31 · ENGINEERING MECHANICS

- A/40 Determine the percent reductions in both area and area moment of inertia
- about the y-axis caused by removal of the rectangular cutout from the rectangular
- plate of baseband height h.

Slide 32 · ENGINEERING MECHANICS

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 33 · ENGINEERING MECHANICS

- A/41 The cross-sectional area of an I-beam has the dimensions shown. Obtain a close
- approximation to the handbook value of Ix = 385(106) mm4 by treating the
- section as being composed of three rectangles.

Slide 34 · ENGINEERING MECHANICS

- A/42 Calculate the moment of inertia of the shaded area about the x-axis.

Slide 35 · ENGINEERING MECHANICS

- A/43 The variable h designates the arbitrary vertical location of the bottom of the
- rectangular cutout within the rectangular area. Determine the area moment of
- inertia about the x-axis for (a) h = 1000 mm and (b) h = 1500 mm.

Slide 36 · ENGINEERING MECHANICS

- A/44 The variable h designates the arbitrary vertical location of the center of the
- circular cutout within the semicircular area. Determine the area moment of inertia
- about the x-axis for (a) h = 0 and (b) h = R/2.

Slide 37 · ENGINEERING MECHANICS

- A/45 Calculate the moment of inertia of the shaded area about the x-axis.

Slide 38 · ENGINEERING MECHANICS

- A/46 Determine the moments of inertia of the shaded area about the x- and y-axes.

Slide 39 · ENGINEERING MECHANICS

- A/48 Determine the moments of inertia of the Z-section about its centroidal xo- and

Slide 40 · ENGINEERING MECHANICS

- A/49 Determine the moment of inertia of the shaded area about the x-axis in two

Slide 41 · P. Ramchandra

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

EV0000210 · EV0000211 · EV0000212 · +120 source units in pack.json

15 Source slide deck: UE20CV101 UNIT 4 SESSION 6

Study focus: Source slide deck: UE20CV101 UNIT 4 SESSION 6. The notes identify each page heading, summarize its main ideas, and explain its place in this unit; citations preserve the source page locator. EV0000402 · EV0000403

SHORT-NOTE RECAP

Study notes · Source slide deck: UE20CV101 UNIT 4 SESSION 6

The slide deck is rewritten here as searchable lesson notes. Each subsection states the slide heading and its concise teaching points; page citations lead back to the indexed source image when visual details are needed.

Slide 1 · ENGINEERING MECHANICS

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 2 · ENGINEERING MECHANICS - STATICS

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 3 · ENGINEERING MECHANICS - STATICS

- Moment of Inertia of a rectangular lamina about its centroidal axis

Slide 4 · ENGINEERING MECHANICS - STATICS

- Moment of Inertia of a semi Circle about its centroidal axis

Slide 5 · ENGINEERING MECHANICS - STATICS

- Moment of Inertia of a Quarter Circle about its centroidal axis

Slide 6 · Rashmi B A

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

EV0000402 · EV0000403 · EV0000404 · +15 source units in pack.json

16 Source slide deck: UE20CV101 UNIT 4 SESSION 11

Study focus: Source slide deck: UE20CV101 UNIT 4 SESSION 11. The notes identify each page heading, summarize its main ideas, and explain its place in this unit; citations preserve the source page locator. EV0000444 · EV0000445

SHORT-NOTE RECAP

Study notes · Source slide deck: UE20CV101 UNIT 4 SESSION 11

The slide deck is rewritten here as searchable lesson notes. Each subsection states the slide heading and its concise teaching points; page citations lead back to the indexed source image when visual details are needed.

Slide 1 · ENGINEERING MECHANICS

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 2 · ENGINEERING MECHANICS - STATICS

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 3 · ENGINEERING MECHANICS - STATICS

- Problem A/44. Determine the moments of inertia of the Z-section
- about its centroidal *xo*- and *yo*-axes.

Slide 4 · ENGINEERING MECHANICS - STATICS

- SOLUTION: Moment of Inertia about centroidal x-axis:
- = MI of rectangle (1) + MI of rectangle (2)
- 80(20)3 20(160)3

Slide 5 · ENGINEERING MECHANICS - STATICS

- SOLUTION: Moment of Inertia about centroidal y -axis:
- = MI of rectangle (1) + MI of rectangle (2)
- 20(80)3 160(20)3

Slide 6 · Rashmi B A

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

EV0000444 · EV0000445 · EV0000446 · +15 source units in pack.json

17 Source slide deck: UE20CV101 UNIT 4 SESSION 12

Study focus: Source slide deck: UE20CV101 UNIT 4 SESSION 12. The notes identify each page heading, summarize its main ideas, and explain its place in this unit; citations preserve the source page locator. EV0000462 · EV0000463

SHORT-NOTE RECAP

Study notes · Source slide deck: UE20CV101 UNIT 4 SESSION 12

The slide deck is rewritten here as searchable lesson notes. Each subsection states the slide heading and its concise teaching points; page citations lead back to the indexed source image when visual details are needed.

Slide 1 · ENGINEERING MECHANICS

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 2 · ENGINEERING MECHANICS - STATICS

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 3 · ENGINEERING MECHANICS - STATICS

- Problem A/60. Calculate the polar radius of gyration of the shaded
- area about its centroid C.

Slide 4 · ENGINEERING MECHANICS - STATICS

- Problem A/60. Calculate the polar radius of gyration of the shaded
- area about its centroid C.

Slide 5 · Rashmi B A

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

EV0000462 · EV0000463 · EV0000464 · +12 source units in pack.json

18 Source slide deck: UE26ME142A Unit 4 class 53 Lecture PPT

Study focus: Source slide deck: UE26ME142A Unit 4 class 53 Lecture PPT. The notes identify each page heading, summarize its main ideas, and explain its place in this unit; citations preserve the source page locator. EV0000477 · EV0000478

SHORT-NOTE RECAP

Study notes · Source slide deck: UE26ME142A Unit 4 class 53 Lecture PPT

The slide deck is rewritten here as searchable lesson notes. Each subsection states the slide heading and its concise teaching points; page citations lead back to the indexed source image when visual details are needed.

Slide 1 · Slide page 1

- no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection.

Slide 2 · Engineering Mechanics

- J. L. Meriam and L. G. Kraige
- Copyright © 2012 by John Wiley & Sons, Inc. All rights reserved.
- S V Venkatesh - Engineering Mechanics – Unit 4 – Chapter 6– Friction

Slide 3 · Engineering Mechanics Statics

- Introduction, Frictional Phenomena - Types of
- Friction, Mechanism of Dry Friction, Static
- S V Venkatesh - Engineering Mechanics – Unit 4 – Chapter 6– Friction

Slide 4 · Engineering Mechanics Statics

- 337 CHAPTER 6 (Text book):
- 6/1, 6/2, 6/3, 6/8
- S V Venkatesh - Engineering Mechanics – Unit 4 – Chapter 6– Friction Numerical problems.

Slide 5 · Engineering Mechanics Statics

- Dry Friction – Unlubricated – two solids
- Fluid Friction – adjacent layers in a
- S V Venkatesh - Engineering Mechanics – Unit 4 – Chapter 6– Friction

Slide 6 · Engineering Mechanics Statics

- Angle of Friction $\tan\alpha = F/N$
- S V Venkatesh - Engineering Mechanics – Unit 4 – Chapter 6– Friction

Slide 7 · Engineering Mechanics Statics

- S V Venkatesh - Engineering Mechanics – Unit 4 – Chapter 6– Friction

Slide 8 · Engineering Mechanics Statics

- The condition of impending motion is known
- to exist. Here a body which is in equilibrium is
- S V Venkatesh - Engineering Mechanics – Unit 4 – Chapter 6– Friction

Slide 9 · Engineering Mechanics Statics

- Neither the condition of impending motion nor the

- condition of motion is known to exist. To determine
- S V Venkatesh - Engineering Mechanics – Unit 4 – Chapter 6– Friction friction force FislessthanthelimitingvalueFmax

Slide 10 · Engineering Mechanics Statics

- impossible, because the surfaces cannot
- The assumption of equilibrium is therefore 10
- S V Venkatesh - Engineering Mechanics – Unit 4 – Chapter 6– Friction

Slide 11 · Engineering Mechanics Statics

- Relative motion is known to exist between
- μkN always gives the kinetic friction force
- S V Venkatesh - Engineering Mechanics – Unit 4 – Chapter 6– Friction

Slide 12 · Engineering Mechanics Statics

- Engineering Mechanics Statics - S V Venkatesh - Department of Civil

Slide 13 · Slide page 13

- no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection.

EV0000477 · EV0000478 · EV0000479 · +32 source units in pack.json

19 Source slide deck: UE20CV101 UNIT 5 SESSION 2

Study focus: Source slide deck: UE20CV101 UNIT 5 SESSION 2. The notes identify each page heading, summarize its main ideas, and explain its place in this unit; citations preserve the source page locator. EV0000512 · EV0000513

SHORT-NOTE RECAP

Study notes · Source slide deck: UE20CV101 UNIT 5 SESSION 2

The slide deck is rewritten here as searchable lesson notes. Each subsection states the slide heading and its concise teaching points; page citations lead back to the indexed source image when visual details are needed.

Slide 1 · ENGINEERING MECHANICS

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 2 · ENGINEERING MECHANICS - STATICS

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 3 · ENGINEERING MECHANICS - STATICS

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 4 · ENGINEERING MECHANICS - STATICS

- Applying the conditions of equilibrium:

Slide 5 · ENGINEERING MECHANICS - STATICS

- Problem 6/1: The force P is applied to the 90-kg crate, which is stationary before
- the force is applied. Determine the magnitude and direction of the friction force
- F exerted by the horizontal surface on the crate if (a) P = 300 N, (b) P = 400 N,

Slide 6 · ENGINEERING MECHANICS - STATICS

- Problem 6/1: Solution (a) When P = 300 N
- W= 90 X 9.81 N
- $\sigma\ Fy = 0\ -90 \times 9.81 + N = 0\ N = 882.9\ N$

Slide 7 · ENGINEERING MECHANICS - STATICS

- Problem 6/1: Solution (b) P = 400 N
- $\sigma\ Fx = 0\ P - F = 0\ F = 400\ N$ W= 90 X 9.81 N
- $\sigma\ Fy = 0\ -90 \times 9.81 + N = 0\ N = 882.9\ N$

Slide 8 · ENGINEERING MECHANICS - STATICS

- Problem 6/1: Solution (c) P = 500 N
- $\sigma\ Fx = 0\ P - F = 0\ F = 500\ N$ W= 90 X 9.81 N
- $Since F > Fmax$, clearly the condition is

Slide 9 · Rashmi B A

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

EV0000512 · EV0000513 · EV0000514 · +24 source units in pack.json

20 Source slide deck: UE20CV101 UNIT 5 SESSION 3

Study focus: Source slide deck: UE20CV101 UNIT 5 SESSION 3. The notes identify each page heading, summarize its main ideas, and explain its place in this unit; citations preserve the source page locator. EV0000563 · EV0000564

SHORT-NOTE RECAP

Study notes · Source slide deck: UE20CV101 UNIT 5 SESSION 3

The slide deck is rewritten here as searchable lesson notes. Each subsection states the slide heading and its concise teaching points; page citations lead back to the indexed source image when visual details are needed.

Slide 1 · ENGINEERING MECHANICS

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 2 · ENGINEERING MECHANICS - STATICS

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 3 · ENGINEERING MECHANICS - STATICS

- Problem 6/3: The force P is applied to the 50-kg block when it is at rest.
- surface on the block if (a) P = 0, (b) P = 200 N, and (c) P = 250 N. (d) What value
- of P is required to initiate motion up the incline? The coefficients of static and

Slide 4 · ENGINEERING MECHANICS - STATICS

- Problem 6/3:Solution (a) when P=0
- $\sigma\ Fy = 0\ -50 \times 9.81 \cos 15 + N = 0$
- The frictional force F = $\mu k N = 0.2\ 473.78$

Slide 5 · ENGINEERING MECHANICS - STATICS

- Problem 6/3:Solution (b) when P=200 N
- $\sigma\ Fy = 0\ -50 \times 9.81 \cos 15 + N + P\sin(20) = 0$
- $-50 \times 9.81 \sin 15\ -F + P\cos(20) = 0\ \sigma\ Fx = 0$

Slide 6 · ENGINEERING MECHANICS - STATICS

- Problem 6/3:Solution (C) when P=250 N
- $\sigma\ Fy = 0\ -50 \times 9.81 \cos 15 + N + P\sin(20) = 0$
- $\sigma\ Fx = 0\ -50 \times 9.81 \sin 15\ -F + P\cos(20) = 0$

Slide 7 · ENGINEERING MECHANICS - STATICS

- Problem 6/3:Solution (d) when P=? Motion up the incline
- $\sigma\ Fy = 0\ -50 \times 9.81 \cos 15 + N + P\sin(20) = 0$
- $\sigma\ Fx = 0\ -50 \times 9.81 \sin 15\ -F + P\cos(20) = 0$

Slide 8 · ENGINEERING MECHANICS - STATICS

- Problem 6/6: Determine the minimum coefficient of static friction μswhich
- will allow the drum with fixed inner hub to be rolled up the 15° incline at a
- steady speed without slipping. What are the corresponding values of the

Slide 9 · ENGINEERING MECHANICS - STATICS

- $\sigma\ 0 + F(r) - Pcos15 = 0\ \mathrm{W}\ Mo = 2$
- $\sigma\ Fx = 0 + F - Wsin15 + P\cos(15) = 0$
- $\sigma\ Fy = 0\ -\mathrm{W} \cos 15 + N + P\sin(15) = 0$

Slide 10 · Rashmi B A

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

EV0000563 · EV0000564 · EV0000565 · +27 source units in pack.json

21 Source slide deck: UE20CV101 UNIT 5 SESSION 6

Study focus: Source slide deck: UE20CV101 UNIT 5 SESSION 6. The notes identify each page heading, summarize its main ideas, and explain its place in this unit; citations preserve the source page locator. EV0000617 · EV0000618

SHORT-NOTE RECAP

Study notes · Source slide deck: UE20CV101 UNIT 5 SESSION 6

The slide deck is rewritten here as searchable lesson notes. Each subsection states the slide heading and its concise teaching points; page citations lead back to the indexed source image when visual details are needed.

Slide 1 · ENGINEERING MECHANICS

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 2 · ENGINEERING MECHANICS - STATICS

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

Slide 3 · ENGINEERING MECHANICS - STATICS

- 6/33 Determine the distance s to which the 90-kg
- painter can climb without causing the 4-m ladder to
- slip at its lower end A. The top of the 15-kg ladder has

Slide 4 · ENGINEERING MECHANICS - STATICS

- $weightoftheladderWl = 15\ X9.81 = 147.15\ N$

Slide 5 · ENGINEERING MECHANICS - STATICS

- 6/36. The homogeneous square body is positioned as shown. If the
- coefficient of static friction at B is 0.40, determine the critical value of the
- angle ϑ below which slipping will occur. Neglect friction at A.

Slide 6 · ENGINEERING MECHANICS - STATICS

- $+\ NA\cos 30 - 0.4\ NB = 0\ s\ s$
- $-NA\cos 30\ s\sin\theta - NA\sin 30\ s\cos\theta = 0$

Slide 7 · Rashmi B A

- Image-only source page; see the visual page locator in `datasets/content-index.json`.

EV0000617 · EV0000618 · EV0000619 · +18 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Official unit catalogue	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/units.json	Official course/unit title and index.
CLASS 46 Introduction to Moment of Inertia transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/046_CLASS_46_Introduction_to_Moment_of_Inertia__f7841a68b028a2d1.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 46 Introduction to Moment of Inertia transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/046_CLASS_46_Introduction_to_Moment_of_Inertia__f7841a68b028a2d1.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 46 Introduction to Moment of Inertia transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/046_CLASS_46_Introduction_to_Moment_of_Inertia__f7841a68b028a2d1.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9988; only the JSON segmentation supplies the displayed timestamps.
CLASS 46 Introduction to Moment of Inertia transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/046_CLASS_46_Introduction_to_Moment_of_Inertia__f7841a68b028a2d1.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 38 Moment of Inertia_ Numerical transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/048_CLASS_38_Moment_of_Inertia__Numerical__77ab2c76fcd84290.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
CLASS 38 Moment of Inertia_ Numerical transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/048_CLASS_38_Moment_of_Inertia__Numerical__77ab2c76fcd84290.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 38 Moment of Inertia_ Numerical transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/048_CLASS_38_Moment_of_Inertia__Numerical__77ab2c76fcd84290.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9985; only the JSON segmentation supplies the displayed timestamps.
CLASS 38 Moment of Inertia_ Numerical transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/048_CLASS_38_Moment_of_Inertia__Numerical__77ab2c76fcd84290.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 39 Moment of Inertia_ Numerical transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/049_CLASS_39_Moment_of_Inertia__Numerical__b51a45df3dcf2238.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 39 Moment of Inertia_ Numerical transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/049_CLASS_39_Moment_of_Inertia__Numerical__b51a45df3dcf2238.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 39 Moment of Inertia_ Numerical transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/049_CLASS_39_Moment_of_Inertia__Numerical__b51a45df3dcf2238.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9984; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
CLASS 39 Moment of Inertia_ Numerical transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/049_CLASS_39_Moment_of_Inertia__Numerical__b51a45df3dcf2238.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 40 Moment of Inertia_ Numerical transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/050_CLASS_40_Moment_of_Inertia__Numerical__0e23de7b877dd375.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 40 Moment of Inertia_ Numerical transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/050_CLASS_40_Moment_of_Inertia__Numerical__0e23de7b877dd375.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 40 Moment of Inertia_ Numerical transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/050_CLASS_40_Moment_of_Inertia__Numerical__0e23de7b877dd375.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9984; only the JSON segmentation supplies the displayed timestamps.
CLASS 40 Moment of Inertia_ Numerical transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/050_CLASS_40_Moment_of_Inertia__Numerical__0e23de7b877dd375.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 41 Moment of Inertia_ Numerical transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/051_CLASS_41_Moment_of_Inertia__Numerical__465b60d43e569bb9.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
CLASS 41 Moment of Inertia_ Numerical transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/051_CLASS_41_Moment_of_Inertia__Numerical__465b60d43e569bb9.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 41 Moment of Inertia_ Numerical transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/051_CLASS_41_Moment_of_Inertia__Numerical__465b60d43e569bb9.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9984; only the JSON segmentation supplies the displayed timestamps.
CLASS 41 Moment of Inertia_ Numerical transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/051_CLASS_41_Moment_of_Inertia__Numerical__465b60d43e569bb9.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 51.1 Moment of Inertia_ Numerical transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/051_CLASS_51.1_Moment_of_Inertia__Numerical__5c991feecd17a895.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 51.1 Moment of Inertia_ Numerical transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/051_CLASS_51.1_Moment_of_Inertia__Numerical__5c991feecd17a895.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 51.1 Moment of Inertia_ Numerical transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/051_CLASS_51.1_Moment_of_Inertia__Numerical__5c991feecd17a895.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9985; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
CLASS 51.1 Moment of Inertia_ Numerical transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/051_CLASS_51.1_Moment_of_Inertia__Numerical__5c991feecd17a895.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 51.2 Moment of Inertia_ Numerical transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/051_CLASS_51.2_Moment_of_Inertia__Numerical__662a876b25afb423.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 51.2 Moment of Inertia_ Numerical transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/051_CLASS_51.2_Moment_of_Inertia__Numerical__662a876b25afb423.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 51.2 Moment of Inertia_ Numerical transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/051_CLASS_51.2_Moment_of_Inertia__Numerical__662a876b25afb423.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9987; only the JSON segmentation supplies the displayed timestamps.
CLASS 51.2 Moment of Inertia_ Numerical transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/051_CLASS_51.2_Moment_of_Inertia__Numerical__662a876b25afb423.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 51.2 Moment of Inertia_ Numerical transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/051_CLASS_51.2_Moment_of_Inertia__Numerical__91dc2d538980f355.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
CLASS 51.2 Moment of Inertia_ Numerical transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/051_CLASS_51.2_Moment_of_Inertia__Numerical__91dc2d538980f355.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 51.2 Moment of Inertia_ Numerical transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/051_CLASS_51.2_Moment_of_Inertia__Numerical__91dc2d538980f355.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9984; only the JSON segmentation supplies the displayed timestamps.
CLASS 51.2 Moment of Inertia_ Numerical transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/051_CLASS_51.2_Moment_of_Inertia__Numerical__91dc2d538980f355.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 47 Friction_ Numerical transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/055_CLASS_47_Friction__Numerical__f6035152d6e2eca8.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 47 Friction_ Numerical transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/055_CLASS_47_Friction__Numerical__f6035152d6e2eca8.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 47 Friction_ Numerical transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/055_CLASS_47_Friction__Numerical__f6035152d6e2eca8.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9992; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
CLASS 47 Friction_ Numerical transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/055_CLASS_47_Friction__Numerical__f6035152d6e2eca8.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 48 Friction_ Numerical transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/056_CLASS_48_Friction__Numerical__6985dd7e23b106ba.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 48 Friction_ Numerical transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/056_CLASS_48_Friction__Numerical__6985dd7e23b106ba.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 48 Friction_ Numerical transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/056_CLASS_48_Friction__Numerical__6985dd7e23b106ba.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9993; only the JSON segmentation supplies the displayed timestamps.
CLASS 48 Friction_ Numerical transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/056_CLASS_48_Friction__Numerical__6985dd7e23b106ba.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 56.1 Friction_ Numerical transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/056_CLASS_56.1_Friction__Numerical__816881a9ab73e7db.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 56.1 Friction_ Numerical transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/056_CLASS_56.1_Friction__Numerical__816881a9ab73e7db.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
CLASS 56.1 Friction_ Numerical transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/056_CLASS_56.1_Friction__Numerical__816881a9ab73e7db.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9991; only the JSON segmentation supplies the displayed timestamps.
CLASS 56.1 Friction_ Numerical transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/056_CLASS_56.1_Friction__Numerical__816881a9ab73e7db.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 56.2 Friction_ Numerical transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/056_CLASS_56.2_Friction__Numerical__59b393edf5066db5.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 56.2 Friction_ Numerical transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/056_CLASS_56.2_Friction__Numerical__59b393edf5066db5.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CLASS 56.2 Friction_ Numerical transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/056_CLASS_56.2_Friction__Numerical__59b393edf5066db5.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9991; only the JSON segmentation supplies the displayed timestamps.
CLASS 56.2 Friction_ Numerical transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/AV_Summaries/056_CLASS_56.2_Friction__Numerical__59b393edf5066db5.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Slide page text: UE26ME142A Unit 4 Class 46 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/046_UE26ME142A_Unit_4_Class_46_Lecture_PPT__d02ef87253dca769.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: UE26ME142A Unit 4 Class 46 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/046_UE26ME142A_Unit_4_Class_46_Lecture_PPT__d02ef87253dca769.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UE26ME142A Unit 4 Class 46 Lecture PPT (converted PDF; original hash ca1b309f28b6be9ea205fed6ee7774af27f38d77c453fa0b5a65e3f2d23a897c)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/046_UE26ME142A_Unit_4_Class_46_Lecture_PPT__d02ef87253dca769.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Moment of Inertia	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Moment of Inertia	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Moment of Inertia (converted PDF; original hash ef55e97507bbf9e623a96a1497d4f5b6794b13107ee60544f779debc752597d6)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/047_Moment_of_Inertia__e26ad086e9c031a8.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE20CV101 UNIT 4 SESSION 7	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/048_UE20CV101_UNIT_4__SESSION-7__f2f50a6e6cd22401.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE20CV101 UNIT 4 SESSION 7	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/048_UE20CV101_UNIT_4__SESSION-7__f2f50a6e6cd22401.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for UE20CV101 UNIT 4 SESSION 7 (converted PDF; original hash 7665256b5e60d1a1be44d07d4530addac006f38fb4733be728f3ce43bd09dded)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/048_UE20CV101_UNIT_4__SESSION-7__f2f50a6e6cd22401.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE20CV101 UNIT 4 SESSION 8	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/049_UE20CV101_UNIT_4__SESSION-8__5fafe633aad8a088.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE20CV101 UNIT 4 SESSION 8	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/049_UE20CV101_UNIT_4__SESSION-8__5fafe633aad8a088.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UE20CV101 UNIT 4 SESSION 8 (converted PDF; original hash 4dfa190774c94f173d3f82dd2e5df5780141f82e2c71c4843c044a9359c5deb1)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/049_UE20CV101_UNIT_4__SESSION-8__5fafe633aad8a088.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE20CV101 UNIT 4 SESSION 9	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/050_UE20CV101_UNIT_4__SESSION-9__9ddd0b077d481238.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE20CV101 UNIT 4 SESSION 9	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/050_UE20CV101_UNIT_4__SESSION-9__9ddd0b077d481238.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for UE20CV101 UNIT 4 SESSION 9 (converted PDF; original hash f14d351c141cd10666496c339542589875fdbada8cc78934218328853 25cba39)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/050_UE20CV101_UNIT_4__SESSION-9__9ddd0b077d481238.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE20CV101 UNIT 4 SESSION 6	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT-4__SESSION-6__f8707bcb8798967a.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE20CV101 UNIT 4 SESSION 6	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT-4__SESSION-6__f8707bcb8798967a.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UE20CV101 UNIT 4 SESSION 6 (converted PDF; original hash d58ed7ece2c1005f874e755008057e82e8920f4ddc0764aa78dba4e65d57fceb)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT-4__SESSION-6__f8707bcb8798967a.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE20CV101 UNIT 4 SESSION 10	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT_4__SESSION-10__12f7ad3eb0e14cdc.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE20CV101 UNIT 4 SESSION 10	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT_4__SESSION-10__12f7ad3eb0e14cdc.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for UE20CV101 UNIT 4 SESSION 10 (converted PDF; original hash 316ff20663c7298b8309dd52cce2181452f92de8de8674062db0522946822a89)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT_4__SESSION-10__12f7ad3eb0e14cdc.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE20CV101 UNIT 4 SESSION 11	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT_4__SESSION-11__5012d0aadf26d705.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE20CV101 UNIT 4 SESSION 11	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT_4__SESSION-11__5012d0aadf26d705.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UE20CV101 UNIT 4 SESSION 11 (converted PDF; original hash a71de0e31f23e83e0787551bd98ca1bd39cb7b21d0448651b154a8e0ea2477a2)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT_4__SESSION-11__5012d0aadf26d705.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE20CV101 UNIT 4 SESSION 12	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT_4__SESSION-12__b705719c21d2c8f4.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE20CV101 UNIT 4 SESSION 12	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT_4__SESSION-12__b705719c21d2c8f4.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for UE20CV101 UNIT 4 SESSION 12 (converted PDF; original hash 9d36774190aac4b8a5689a0965af3aea3acd2a27165f4abaec9a21a209ce4fc0)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/051_UE20CV101_UNIT_4__SESSION-12__b705719c21d2c8f4.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE26ME142A Unit 4 class 53 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/053_UE26ME142A_Unit_4_class_53-Lecture_PPT__980f9dd97d5e9157.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE26ME142A Unit 4 class 53 Lecture PPT	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/053_UE26ME142A_Unit_4_class_53-Lecture_PPT__980f9dd97d5e9157.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UE26ME142A Unit 4 class 53 Lecture PPT (converted PDF; original hash 742906e1e1b2d3d2534379c883f30a16d6e881ca1044e876a4042effa759d096)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/053_UE26ME142A_Unit_4_class_53-Lecture_PPT__980f9dd97d5e9157.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE20CV101 UNIT 5 SESSION 2	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/054_UE20CV101_UNIT-5__SESSION-2__65d202baa1c9be91.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE20CV101 UNIT 5 SESSION 2	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/054_UE20CV101_UNIT-5__SESSION-2__65d202baa1c9be91.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for UE20CV101 UNIT 5 SESSION 2 (converted PDF; original hash 434a3671d2282b9341027f0a6bf8b159030c3daa324760d5cd72fbd4690b1dc6)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/054_UE20CV101_UNIT-5__SESSION-2__65d202baa1c9be91.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE20CV101 UNIT 5 SESSION 4	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/055_UE20CV101_UNIT-5__SESSION-4__e7632168d688ab28.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE20CV101 UNIT 5 SESSION 4	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/055_UE20CV101_UNIT-5__SESSION-4__e7632168d688ab28.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UE20CV101 UNIT 5 SESSION 4 (converted PDF; original hash defd460c19a5a7a0156848f7216b87686028436758cec4aba7f952dee1e326ff)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/055_UE20CV101_UNIT-5__SESSION-4__e7632168d688ab28.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE20CV101 UNIT 5 SESSION 3	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/056_UE20CV101_UNIT-5__SESSION-3__f5f42ae0569674bb.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE20CV101 UNIT 5 SESSION 3	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/056_UE20CV101_UNIT-5__SESSION-3__f5f42ae0569674bb.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for UE20CV101 UNIT 5 SESSION 3 (converted PDF; original hash 9e8880bf871d0e58a2b76c25bf1abdcb842bb6ee1c4e5ef947e5ed3f2c7b17fd)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/056_UE20CV101_UNIT-5__SESSION-3__f5f42ae0569674bb.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE20CV101 UNIT 5 SESSION 5	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/056_UE20CV101_UNIT-5__SESSION-5__03f6ff4c881c01b2.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE20CV101 UNIT 5 SESSION 5	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/056_UE20CV101_UNIT-5__SESSION-5__03f6ff4c881c01b2.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for UE20CV101 UNIT 5 SESSION 5 (converted PDF; original hash d128c486eada7966fd32e9a673d62895bae54293b5a6b58e496a12c2ff11e6d7)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/056_UE20CV101_UNIT-5__SESSION-5__03f6ff4c881c01b2.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: UE20CV101 UNIT 5 SESSION 6	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/056_UE20CV101_UNIT-5__SESSION-6__a12f1678f32da358.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: UE20CV101 UNIT 5 SESSION 6	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/056_UE20CV101_UNIT-5__SESSION-6__a12f1678f32da358.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for UE20CV101 UNIT 5 SESSION 6 (converted PDF; original hash 23913070dbdd60818ee3f8223fa075a73f2028509abc798885d8c546f6162a45)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26ME142A_Engineering_Mechanics_1/Unit_04/Slides/056_UE20CV101_UNIT-5__SESSION-6__a12f1678f32da358.pdf	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 2 rendered pages in 046_UE26ME142A_Unit_4_Class_46_Lecture_PPT__d02ef87253dca769.slides.md have no readable `.slides.md` text; their original page images are included.
2. 2 rendered pages in 053_UE26ME142A_Unit_4_class_53-Lecture_PPT__980f9dd97d5e9157.slides.md have no readable `.slides.md` text; their original page images are included.
3. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
4. 4 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
5. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
6. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
7. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
8. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00001	1 covered · 0 excluded · 0 gaps	catalogue: Unit title and index copied from the official units.json catalogue; human content review is pending.
SRC00002	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0064. Sidecar variants compare...
SRC00003	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0064. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00004	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0064. Sidecar variants compare...
SRC00005	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0064. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00006	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0059. Sidecar variants compare...
SRC00007	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0059. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00008	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0059. Sidecar variants compare...
SRC00009	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0059. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00010	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0051. Sidecar variants compare...
SRC00011	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0051. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00012	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0051. Sidecar variants compare...
SRC00013	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0051. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00014	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0065. Sidecar variants compare...
SRC00015	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0065. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00016	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0065. Sidecar variants compare...
SRC00017	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0065. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00018	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0023. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0024 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0063. Sidecar variants compare...
SRC00019	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0023. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0024 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0063. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00020	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0023. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0024 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0063. Sidecar variants compare...
SRC00021	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0023. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0024 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0063. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00022	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0023. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0024 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0063. Sidecar variants compare...
SRC00023	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0023. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0024 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0063. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00024	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0023. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0024 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0063. Sidecar variants compare...
SRC00025	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0023. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0024 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0063. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00026	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0057. Sidecar variants compare...
SRC00027	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0057. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00028	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0057. Sidecar variants compare...
SRC00029	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0038. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0039 to s0057. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00030	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0061. Sidecar variants compare...
SRC00031	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0061. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00032	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0061. Sidecar variants compare...
SRC00033	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0061. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00034	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare...
SRC00035	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00036	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare...
SRC00037	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0042 to s0061. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00038	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare...
SRC00039	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00040	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare...
SRC00041	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00042	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0059. Sidecar variants compare...
SRC00043	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0059. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00044	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0059. Sidecar variants compare...
SRC00045	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0041 to s0059. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00046	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0062. Sidecar variants compare...
SRC00047	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0062. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00048	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0062. Sidecar variants compare...
SRC00049	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0022 to s0043. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0044 to s0062. Sidecar variants compare...
SRC00050	0 covered · 0 excluded · 6 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...

Source	Unit counts	Example decisions
SRC00052	0 covered · 0 excluded · 8 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00051	0 covered · 0 excluded · 6 gaps	p1: Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page;... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00053	41 covered · 0 excluded · 0 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00055	41 covered · 0 excluded · 0 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...

Source	Unit counts	Example decisions
SRC00054	41 covered · 0 excluded · 0 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00056	0 covered · 0 excluded · 7 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00058	0 covered · 0 excluded · 7 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00057	0 covered · 0 excluded · 7 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...

Source	Unit counts	Example decisions
SRC00059	0 covered · 0 excluded · 7 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00061	0 covered · 0 excluded · 7 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00060	0 covered · 0 excluded · 7 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00062	0 covered · 0 excluded · 9 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00064	0 covered · 0 excluded · 9 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...

Source	Unit counts	Example decisions
SRC00063	0 covered · 0 excluded · 9 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00065	6 covered · 0 excluded · 0 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00067	6 covered · 0 excluded · 0 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00066	6 covered · 0 excluded · 0 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...

Source	Unit counts	Example decisions
SRC00068	0 covered · 0 excluded · 8 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00070	0 covered · 0 excluded · 8 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00069	0 covered · 0 excluded · 8 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00071	6 covered · 0 excluded · 0 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00073	6 covered · 0 excluded · 0 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...

Source	Unit counts	Example decisions
SRC00072	6 covered · 0 excluded · 0 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00074	5 covered · 0 excluded · 0 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00076	5 covered · 0 excluded · 0 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00075	5 covered · 0 excluded · 0 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...

Source	Unit counts	Example decisions
SRC00077	11 covered · 0 excluded · 0 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00079	13 covered · 0 excluded · 0 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00078	11 covered · 0 excluded · 0 gaps	p1: Alternate extracted slide text page 1; text differs from or is absent in the .slides.md page;... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00080	9 covered · 0 excluded · 0 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00082	9 covered · 0 excluded · 0 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...

Source	Unit counts	Example decisions
SRC00081	9 covered · 0 excluded · 0 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00083	0 covered · 0 excluded · 8 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00085	0 covered · 0 excluded · 8 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00084	0 covered · 0 excluded · 8 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...

Source	Unit counts	Example decisions
SRC00086	10 covered · 0 excluded · 0 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00088	10 covered · 0 excluded · 0 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00087	10 covered · 0 excluded · 0 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00089	0 covered · 0 excluded · 8 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00091	0 covered · 0 excluded · 8 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...

Source	Unit counts	Example decisions
SRC00090	0 covered · 0 excluded · 8 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...
SRC00092	7 covered · 0 excluded · 0 gaps	p1: Extracted slide page 1 from... p2: Extracted slide page 2 from... p3: Extracted slide page 3 from...
SRC00094	7 covered · 0 excluded · 0 gaps	p1: Original slide page 1 rendered from... p2: Original slide page 2 rendered from... p3: Original slide page 3 rendered from...
SRC00093	7 covered · 0 excluded · 0 gaps	p1: Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human... p2: Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human... p3: Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human...

C Revision and retrieval practice